ExtraaLearn Data Analysis

PGP DSBA - Guided Project Report

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Contents

1. Description: - Machine Learning 1: ExtraaLearn Data Analysis: - 6
2. Data Description 6
2.1 Data Dictionary 6
3. Data Overview 8
3.1 Displaying the first five rows of the dataset: - 8
3.2 View the last 5 rows of the dataset: - 8
3.3 Checking the shape of the dataset: - 8
3.4 Check the data types of the columns for the dataset. 8
3.5 Checking the duplicate values 8
3.6 Checking the number of unique values 8
3.7 Checking the statistical summary 9
3.8 List of all categorical variables: - 9
Exploratory Data Analysis 10
4.0 Univariate Analysis: - 10
4.1 Observations on age 10
4.2 Observations on website_visits: - 11
4.7 Observations on profile_completed 11
4.3 Observations on number of time_spent_on_website 12
4.4 Observations on number of page_views_per_visit 13
4.5 Observations on number of adults: - 13
4.6 Observations on the number of children 14
4.8 Observations on last_activity 14
4.9 Observations on print_media_type1 14
4.10 Observations on print_media_type2 15
4.11 Observations on room type reserved 15
4.12 Observations on educational_channels 16
4.13 Observations on referral 16
5.0 Bivariate Analysis 17
5.1 Leads will have different expectations from the outcome of the course and the current occupation may play a key role for them to take the program. Let's analyze it. 18
5.2 Boxplot between current_occupation and age 18
5.3 Group by Current occupation: - 19
5.4 Time spent on website V/s Density: - 19
5.5 checking the median value: - 20
5.6 To get distribution_plot_wrt_target for website_visits and status: - 20

5.7 To get distribution_plot_wrt_target for page_views_per_visit and status: - 20
5.8 To get stacked_barplot for profile_completed and status 21
5.9 To get stacked_barplot for last_activity and status..... 22
5.10 To get stacked_barplot between print_media_type1 and status 22
5.11 To get stacked_barplot between print_media_type1 and status 23
5.12 To get stacked_barplot between digital_media and status: - 23
5.13 To get stacked_barplot between educational_channels and status 24
5.14 To get stacked_barplot between referral and status 24
6.0 Data Preprocessing 25
6.1 Outlier Check: - Let's check for outliers in the data. 25
6.2 Data Preparation for modeling 25
7.0 Model Building 26
7.1 Logistic Regression (with Statsmodel)..... 26
7.1.1 Checking Logistic Regression model performance on training set 27
7.1.2 Checking Logistic Regression model performance on test set..... 27
7.2 Naive - Bayes Classifier..... 28
7.2.1 Build Naive Bayes Model..... 28
7.2.2 Checking Naive - Bayes Classifier performance on training set 28
7.2.3 Checking Naive - Bayes Classifier performance on test set 29
7.3 KNN Classifier (K = 3)..... 29
7.3.1 Checking KNN Classifier performance on training set 30
7.3.2 Checking KNN Classifier performance on test set..... 30
7.4 Decision Tree Classifier..... 31
7.4.1 Checking Decision Tree Classifier performance on training set 31
7.4.2 To get model performance on test data..... 31
8.0 Model Performance Improvement 32
9.0 Training the Logistic Regression model again with only the significant features..... 33
9.1 Determining optimal threshold using ROC Curve 34
9.3 Find the optimal threshold 34
9.4 Checking tuned Logistic Regression model performance on test set 35
10.0 KNN Classifier (different values of K) 36
10.1 To build KNN model with number of neighbors as best_k 37
10.2 Checking tuned KNN model performance on training set 37
10.3 Checking tuned KNN model performance on test set 37
11.0 Decision Tree Classifier (pre-pruning)..... 38
11.1 Checking tuned Decision Tree Classifier performance on training set 38

11.2 Checking tuned Decision Tree Classifier performance on test set 39
12.0 Visualizing the Decision Tree 39
13.0 Model Performance Comparison and Final Model Selection 40
14.0 Actionable Insights and Recommendations 40
14.1 Actionable Insights: - 40
14.2 Recommendations: - 40

List of Tables: -

Table 1 Top 5 Rows 8
Table 2 Last 5 rows 8
Table 3 Data Columns type 8
Table 4 Statistical summary 9
Table 5 Categorical variables 10
Table 6 Group by current occupation 19
Table 7 Median value for time spent on website 20
Table 8 Status wise Profile completed 21
Table 9 status wise last activity 22
Table 10 status wise print media type1 22
Table 11 status wise print media type2 23
Table 12 status wise digital media count 23
Table 13 status wise educational channel count 24
Table 14 status wise count for referral 25
Table 15 split data into train & test ratio 75:25 26
Table 16 Logistic Regression Model Building 26
Table 17 Logistic regression model performance training set 27
Table 18 Logistic regression model performance test set 27
Table 19 Naive - Bayes Classifier performance on training set 28
Table 20 Naive - Bayes Classifier performance on test set 29
Table 21 KNN Classifier performance on training set 30
Table 22 KNN Classifier performance on test set 30
Table 23 Decision Tree Classifier performance on training set 31
Table 24 Decision Tree Classifier performance on test set 32
Table 25 Calculate VIF 32
Table 26 High p-value variables 33
Table 27 test data with selected features 33
Table 28 Logistic Regression model again with only the significant features 34
Table 29 tuned Logistic Regression model performance on training set 35
Table 30 tuned Logistic Regression model performance on test set 35
Table 31 best k value and its recall score 36
Table 32 tuned KNN model performance on training set 37
Table 33 tuned KNN model performance on test set 37
Table 34 tuned Decision Tree Classifier performance on training set 38
Table 35 tuned Decision Tree Classifier performance on test set 39
Table 36 Training performance comparison: 40
Table 37 Test set performance comparison 40

List of Figures: -

Figure 1 Observations on age 11
Figure 2 Observations on website_visits 11
Figure 7 Observations on profile_completed 12
Figure 3 Observations on number of time_spent_on_website 12
Figure 4 Observations on number of page_views_per_visit 13
Figure 5 Observations on number of adults 13
Figure 6 Observations on the number of children 14
Figure 8 Observations on last_activity 14
Figure 9 Observations on print_media_type1 15
Figure 10 Observations on print_media_type2 15
Figure 11 Observations on room type reserved 16
Figure 12 Observations on educational_channels 16
Figure 13 Observations on referral 17
Figure 14 Heatmap to view the correlation between parameters 17
Figure 15 current_occupation & status 18
Figure 16 current_occupation and age 18
Figure 17 Time spent on website V/s Density 19
Figure 18 distribution_plot_wrt_target for website_visits and status 20
Figure 19 distribution_plot_wrt_target for page_views_per_visit and status 21
Figure 20 stacked_barplot for profile_completed and status 22
Figure 21 stacked_barplot for last_activity and status 22
Figure 22 stacked_barplot between print_media_type1 and status 23
Figure 23 stacked_barplot between print_media_type1 and status 23
Figure 24 stacked_barplot between digital_media and status 24
Figure 25 stacked_barplot between educational_channels and status 24
Figure 26 stacked_barplot between referral and status 25
Figure 27 Outlier Check 25
Figure 28 confusion_matrix training data logistic regression 27
Figure 29 Logistic Regression model performance on test set 28
Figure 30 Naive - Bayes Classifier performance on training set 29
Figure 31 Naive - Bayes Classifier performance on training set confusion matrix 29
Figure 32 KNN Classifier (K = 3) confusion matrix training data 30
Figure 33 KNN Classifier (K = 3) confusion matrix test data 31
Figure 34 Decision Tree Classifier performance on training set confusion matrix 31
Figure 35 Decision Tree Classifier performance on test set confusion matrix 32
Figure 36 optimal threshold using ROC Curve 34
Figure 37 Logistic Regression model performance on training set confusion matrix 35
Figure 38 tuned Logistic Regression model performance on test set confusion matrix 35
Figure 39 tuned KNN model performance on training set confusion matrix 37
Figure 40 tuned KNN model performance on test set confusion matrix 38
Figure 41 tuned Decision Tree Classifier performance on training set confusion matrix 38
Figure 42 tuned Decision Tree Classifier performance on test set confusion matrix 39
Figure 43 Visualizing the Decision Tree 39

1. Description:- Machine Learning 1: ExtraaLearn Data Analysis:-

Business Context

The EdTech industry has been surging in the past decade immensely, and according to a forecast, the Online Education market would be worth $286.62bn by 2023 with a compound annual growth rate (CAGR) of 10.26% from 2018 to 2023. The modern era of online education has enforced a lot in its growth and expansion beyond any limit. Due to having many dominant features like ease of information sharing, personalized learning experience, transparency of assessment, etc, it is now preferable to traditional education.

In the present scenario due to the Covid-19, the online education sector has witnessed rapid growth and is attracting a lot of new customers. Due to this rapid growth, many new companies have emerged in this industry. With the availability and ease of use of digital marketing resources, companies can reach out to a wider audience with their offerings. The customers who show interest in these offerings are termed as leads. There are various sources of obtaining leads for Edtech companies, like

- The customer interacts with the marketing front on social media or other online platforms.
- The customer browses the website/app and downloads the brochure
- The customer connects through emails for more information.

The company then nurtures these leads and tries to convert them to paid customers. For this, the representative from the organization connects with the lead on call or through email to share further details.

Objective

ExtraaLearn is an initial stage startup that offers programs on cutting-edge technologies to students and professionals to help them upskill/reskill. With a large number of leads being generated on a regular basis, one of the issues faced by ExtraaLearn is to identify which of the leads are more likely to convert so that they can allocate resources accordingly. You, as a data scientist at ExtraaLearn, have been provided the leads data to:

- Analyze and build an ML model to help identify which leads are more likely to convert to paid customers,
- Find the factors driving the lead conversion process
- Create a profile of the leads which are likely to convert

2. Data Description

The data contains the different attributes of leads and their interaction details with ExtraaLearn. The detailed data dictionary is given below.

2.1 Data Dictionary

- ID: ID of the lead.
- age: Age of the lead.
- current_occupation: Current occupation of the lead. Values include 'Professional','Unemployed',and 'Student'.

- first_interaction: How did the lead first interacted with ExtraaLearn. Values include 'Website', 'Mobile App'.
- profile_completed: What percentage of profile has been filled by the lead on the website/mobile app. Values include Low - (0-50%), Medium - (50-75%), High (75-100%).
- website_visits: How many times has a lead visited the website.
- time_spent_on_website: Total time spent on the website in seconds.
- page_views_per_visit: Average number of pages on the website viewed during the visits.
- last_activity: Last interaction between the lead and ExtraaLearn.
    - Email Activity: Seeking for details about program through email, Representative shared information with lead like brochure of program , etc.
    - Phone Activity: Had a Phone Conversation with representative, Had conversation over SMS with representative, etc.
    - Website Activity: Interacted on live chat with representative, Updated profile on website, etc.
- print_media_type1: Flag indicating whether the lead had seen the ad of ExtraaLearn in the Newspaper.
- print_media_type2: Flag indicating whether the lead had seen the ad of ExtraaLearn in the Magazine.
- digital_media: Flag indicating whether the lead had seen the ad of ExtraaLearn on the digital platforms.
- educational_channels: Flag indicating whether the lead had heard about ExtraaLearn in the education channels like online forums, discussion threads, educational websites, etc.
- referral: Flag indicating whether the lead had heard about ExtraaLearn through reference.
- status: Flag indicating whether the lead was converted to a paid customer or not.

3. Data Overview

3.1 Displaying the first five rows of the dataset:-

- The data frame has 15 columns as mentioned in the Data Dictionary.
- Data in each row corresponds to the leads and their interaction details with ExtraaLearn.

	ID	age	current_occupation	first_interaction	profile_completed	website_visits	time_spent_on_website	page_views_per_visit	last_activity	print_media_type1	print_media_type2	digital_media	educational_channels	referral	status
0	EXT001	57	Unemployed	Website	High	7	1639	1.86100	Website Activity	Yes	No	Yes	No	No	1
1	EXT002	56	Professional	Mobile App	Medium	2	83	0.32000	Website Activity	No	No	No	Yes	No	0
2	EXT003	52	Professional	Website	Medium	3	330	0.07400	Website Activity	No	No	Yes	No	No	0
3	EXT004	53	Unemployed	Website	High	4	464	2.05700	Website Activity	No	No	No	No	No	1
4	EXT005	23	Student	Website	High	4	600	16.91400	Email Activity	No	No	No	No	No	0

Table 1 Top 5 Rows

3.2 View the last 5 rows of the dataset:-

	ID	age	current_occupation	first_interaction	profile_completed	website_visits	time_spent_on_website	page_views_per_visit	last_activity	print_media_type1	print_media_type2	digital_media	educational_channels	referral	status
4607	EXT4608	35	Unemployed	Mobile App	Medium	15	360	2.17000	Phone Activity	No	No	No	Yes	No	0
4608	EXT4609	55	Professional	Mobile App	Medium	8	2327	5.39300	Email Activity	No	No	No	No	No	0
4609	EXT4610	58	Professional	Website	High	2	212	2.69200	Email Activity	No	No	No	No	No	1
4610	EXT4611	57	Professional	Mobile App	Medium	1	154	3.87900	Website Activity	Yes	No	No	No	No	0
4611	EXT4612	55	Professional	Website	Medium	4	2290	2.07500	Phone Activity	No	No	No	No	No	0

Table 2 Last 5 rows

3.3 Checking the shape of the dataset:-

- The dataset contains information (15 attributes) about 4612 ID's of the lead.

3.4 Check the data types of the columns for the dataset.

- The column range index has 4612 entries (i.e No.of rows = 4612) which starts the number series from 0 to 4611.

```
RangeIndex: 4612 entries, 0 to 4611
Data columns (total 15 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   ID                     4612 non-null   object
 1   age                    4612 non-null   int64
 2   current_occupation     4612 non-null   object
 3   first_interaction      4612 non-null   object
 4   profile_completed      4612 non-null   object
 5   website_visits         4612 non-null   int64
 6   time_spent_on_website  4612 non-null   int64
 7   page_views_per_visit   4612 non-null   float64
 8   last_activity          4612 non-null   object
 9   print_media_type1      4612 non-null   object
 10  print_media_type2      4612 non-null   object
 11  digital_media          4612 non-null   object
 12  educational_channels   4612 non-null   object
 13  referral               4612 non-null   object
 14  status                 4612 non-null   int64
dtypes: float64(1), int64(4), object(10)
memory usage: 540.6+ KB
```

Table 3 Data Columns type

3.5 Checking the duplicate values

- There are 0 (i.e. zero) duplicate values.

3.6 Checking the number of unique values

- The number of unique values is 4612.

3.7 Checking the statistical summary

- This table reflects the statistical summary of the data contains - Count, Mean, Std (Standard deviation), Min (Minimum), Max (Maximum), and Percentages (25%, 50%, 75%).

	age	website_visits	time_spent_on_website	page_views_per_visit	status
count	4612.00000	4612.00000	4612.00000	4612.00000	4612.00000
mean	46.20121	3.56678	724.01127	3.02613	0.29857
std	13.16145	2.82913	743.82868	1.96812	0.45768
min	18.00000	0.00000	0.00000	0.00000	0.00000
25%	36.00000	2.00000	148.75000	2.07775	0.00000
50%	51.00000	3.00000	376.00000	2.79200	0.00000
75%	57.00000	5.00000	1336.75000	3.75625	1.00000
max	63.00000	30.00000	2537.00000	18.43400	1.00000

Table 4 Statistical summary

3.8 List of all categorical variables:-

The number of count of unique values in each column

```
current_occupation
Professional    2616
Unemployed      1441
Student          555
Name: count, dtype: int64
--------------------------------------------------
first_interaction
Website       2542
Mobile App    2070
Name: count, dtype: int64
--------------------------------------------------
profile_completed
High      2264
Medium    2241
Low        107
Name: count, dtype: int64
--------------------------------------------------
last_activity
Email Activity      2278
Phone Activity      1234
Website Activity    1100
Name: count, dtype: int64
--------------------------------------------------
print_media_type1
No     4115
Yes     497
Name: count, dtype: int64
--------------------------------------------------
print_media_type2
No     4379
Yes     233
Name: count, dtype: int64
--------------------------------------------------
digital_media
No     4085
Yes     527
Name: count, dtype: int64
--------------------------------------------------
educational_channels
No     3907
Yes     705
Name: count, dtype: int64
--------------------------------------------------
referral
No     4519
Yes      93
Name: count, dtype: int64
--------------------------------------------------
```

Table 5 Categorical variables

Exploratory Data Analysis

- EDA has a key role in data analysis for the data science. It includes different types of analysis like Univariate analysis & Bivariate analysis which are explained in the below points.

4.0 Univariate Analysis:-

It means one variable analysis with count.

4.1 Observations on age

It is to create a histogram_boxplot for 'Observations on age'.

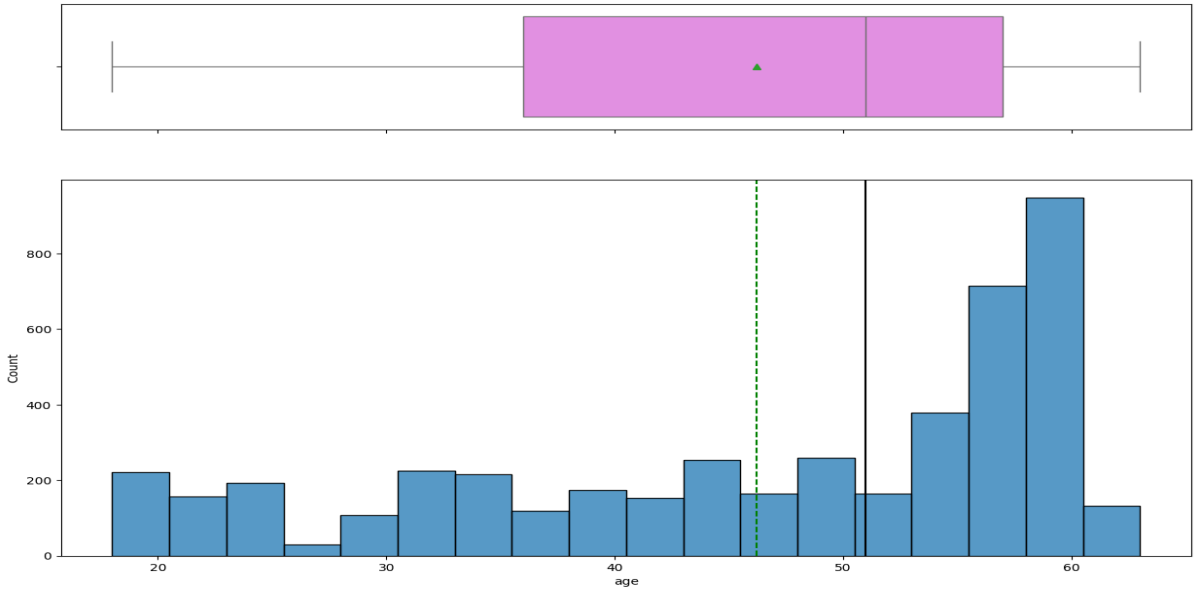

Figure 1 Observations on age

- The Observations on age are distributed approximately 46.20 (Approximately) by leads on average.
- It has no outliers and it is left skewed.
- Median is 51.

4.2 Observations on website_visits:-

It is to create a histogram_boxplot for 'website_visits'.

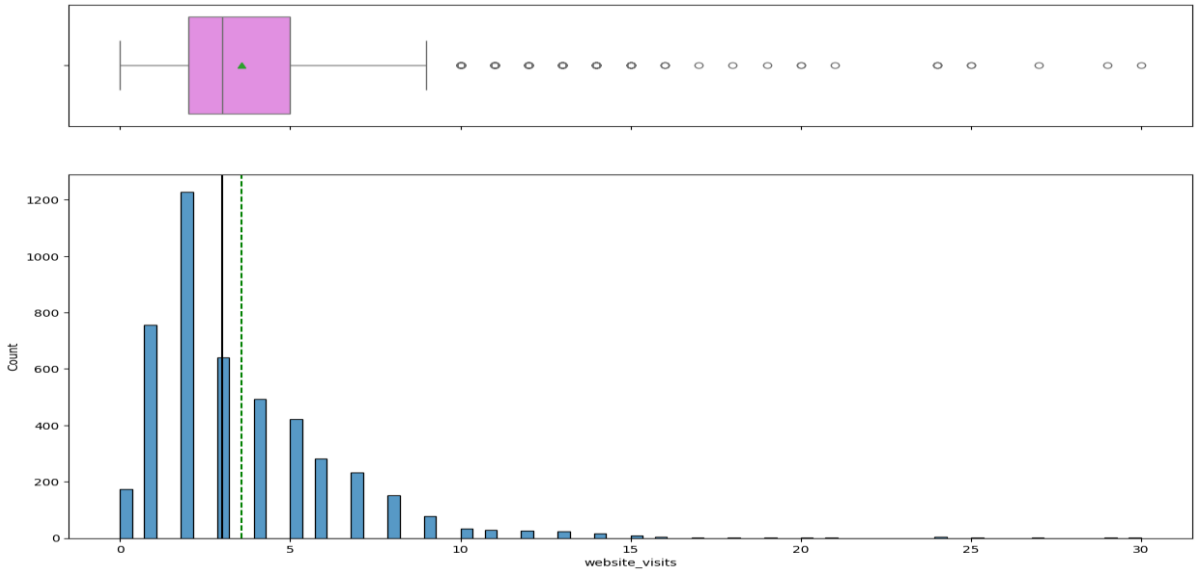

Figure 2 Observations on website_visits

- The Observations on website_visits are distributed approximately 3.56 by leads on average.
- It has many outliers and it is right skewed.
- Median is 3.

4.7 Observations on profile_completed

The bar chart reflects 3 types of profiles completed.

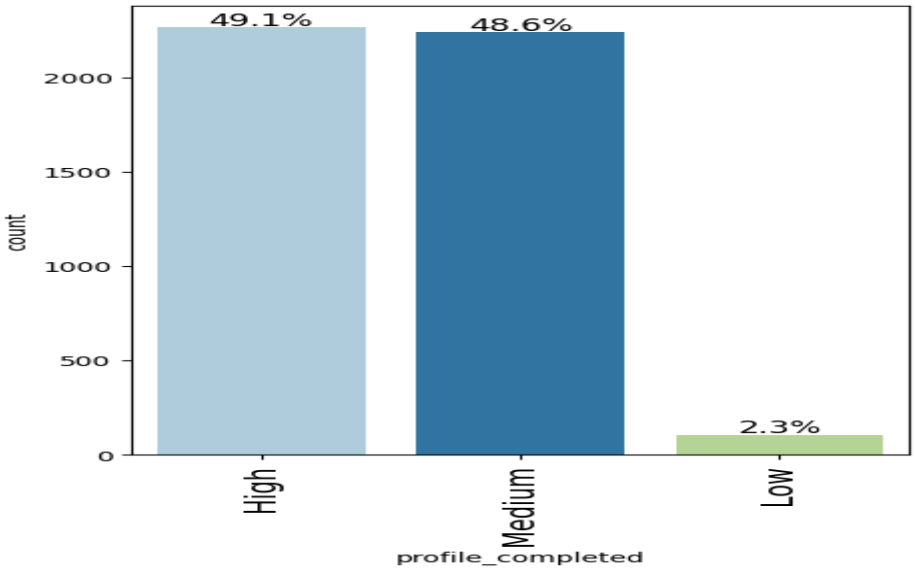

Figure 3 Observations on profile_completed

49.1% - A high percentage of profile has been filled by the lead on the website/mobile app.

48.6% - Medium percentage of profile has been filled by the lead on the website/mobile app.

2.3% - A low percentage of profile has been filled by the lead on the website/mobile app.

4.3 Observations on number of time_spent_on_website

It is to create a histogram_boxplot for 'number of time_spent_on_website'.

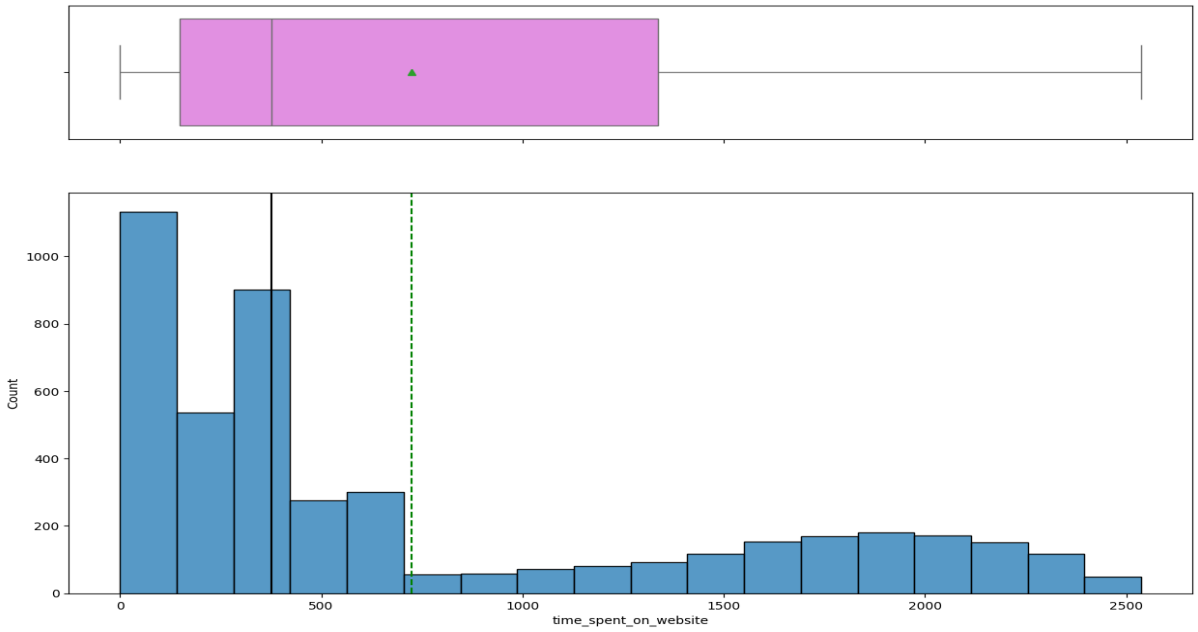

Figure 4 Observations on number of time_spent_on_website

- The Observations on time_spent_on_website are distributed approximately 724.01 by leads on average.
- It has no outliers and it is right skewed.
- Median is 376.

4.4 Observations on number of page_views_per_visit

It is to create a histogram_boxplot for 'number of page_views_per_visit'.

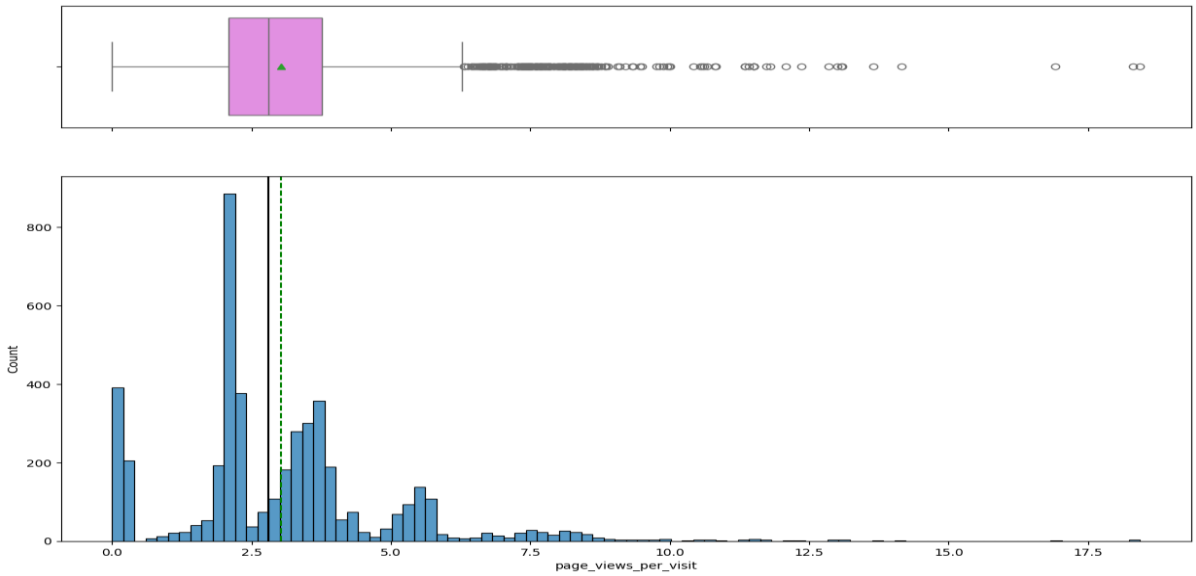

Figure 5 Observations on number of page_views_per_visit

- The Observations on page_views_per_visit are distributed approximately 3.02 by leads on average.
- It has many outliers to right hand side and it is right skewed.
- Median is 2.79 on page_views_per_visit.

4.5 Observations on number of adults:-

The bar chart reflects 3 types of current occupations.

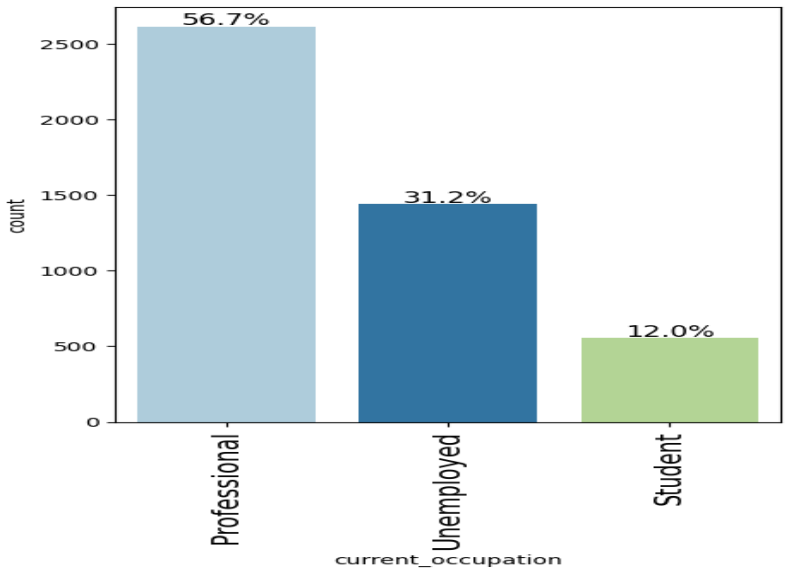

Figure 6 Observations on number of adults

- It gradually slopes down to the right-hand side.
- Current occupation of the leads for Professional is 56.7%, Unemployed is 31.2% and Students with 12.0%.

4.6 Observations on the number of children

The bar chart reflects a number of children with various categories of age.

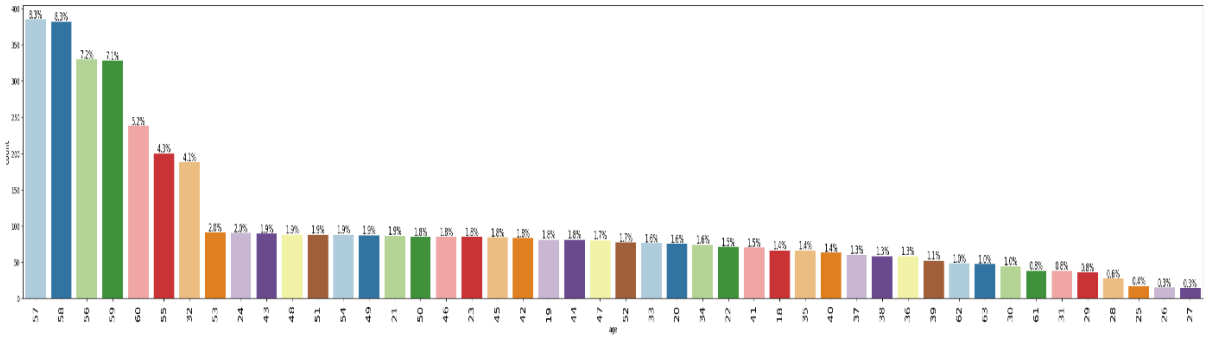

Figure 7 Observations on the number of children

- The 57-58 age group occupied 8.3% which is the highest one.
- 0.3% of the children's age group is 26-27, the least.
- It has gradually reduced from 8.3% to 2.0% and slowly reduced with a small change.

4.8 Observations on last_activity

The bar chart reflects 3 types of unique values for last_activity.

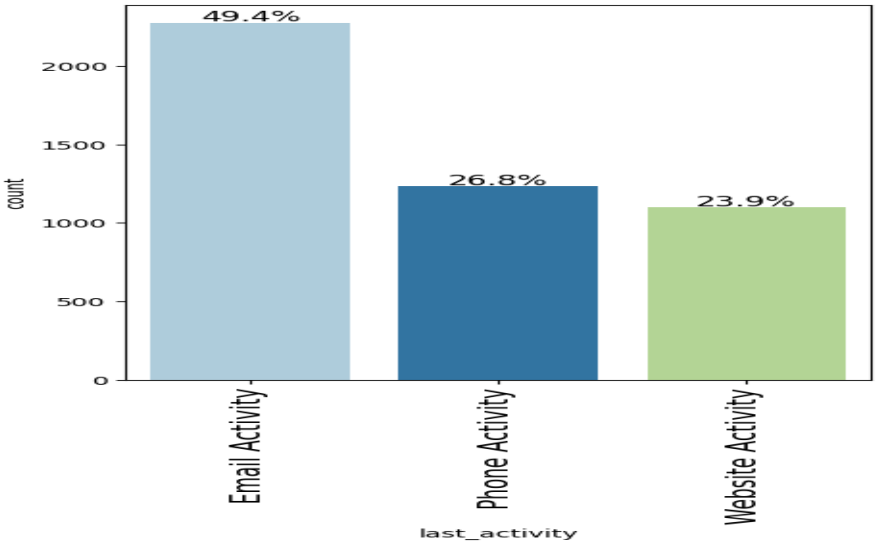

Figure 8 Observations on last_activity

The last interaction between the lead and ExtraaLearn for Email Activity is 49.4%.

The last interaction between the lead and ExtraaLearn for Phone Activity is 26.8%.

The last interaction between the lead and ExtraaLearn for Website Activity is 23.9%.

4.9 Observations on print_media_type1

The bar chart reflects Yes or No percentages for print_media_type1.

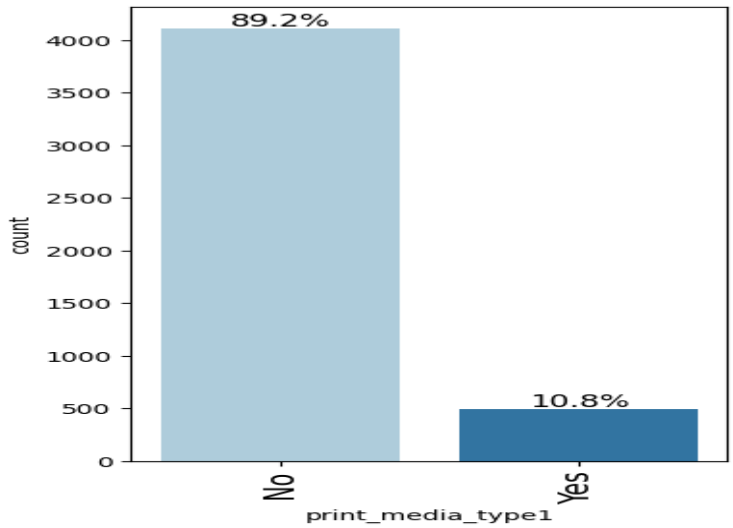

Figure 9 Observations on print_media_type1

No lead had seen the ad of ExtraaLearn in the Newspaper is 89.2%.

Yes lead had seen the ad of ExtraaLearn in the Newspaper is 10.8%.

4.10 Observations on print_media_type2

The bar chart reflects Yes or No percentages for print_media_type2.

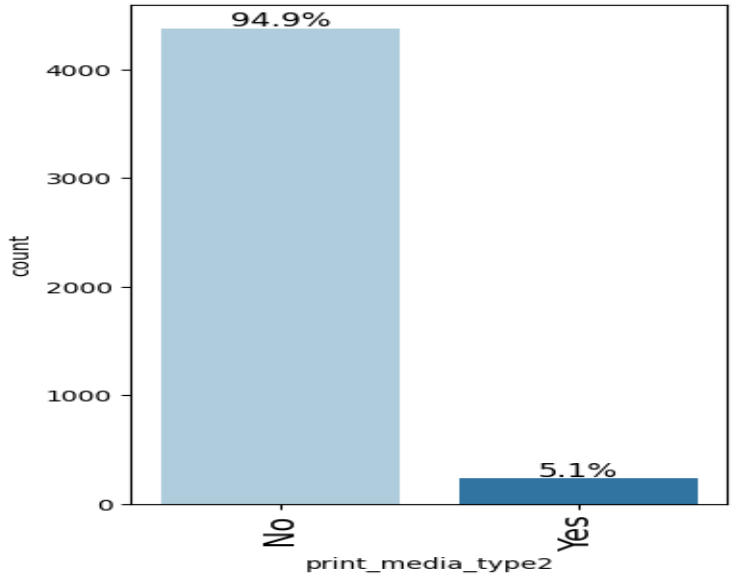

Figure 10 Observations on print_media_type2

No lead had seen the ad of ExtraaLearn in the Magazine is 94.9%.

Yes lead had seen the ad of ExtraaLearn in the Magazine is 5.1%.

4.11 Observations on room type reserved

The bar chart reflects Yes or No percentages for digital_media.

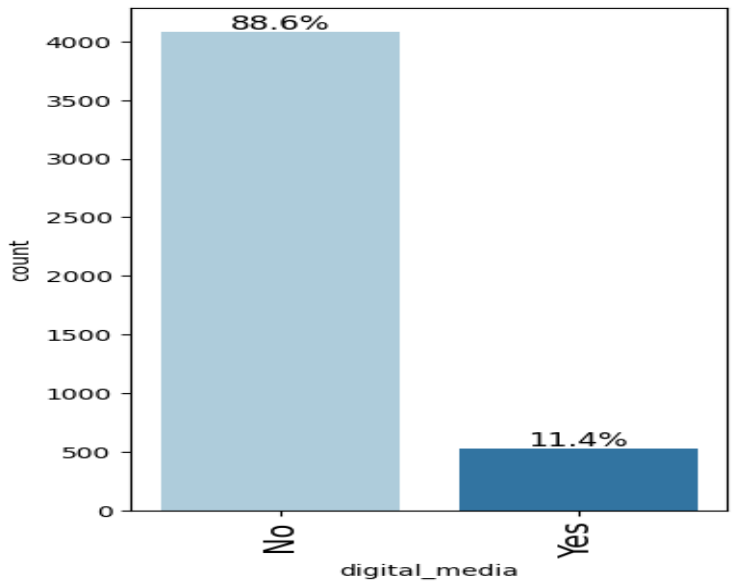

Figure 11 Observations on room type reserved

No lead had seen the ad of ExtraaLearn on the digital platforms is 88.6%.

Yes lead had seen the ad of ExtraaLearn on the digital platforms is 11.4%.

4.12 Observations on educational_channels

The bar chart reflects Yes or No percentages for educational_channels.

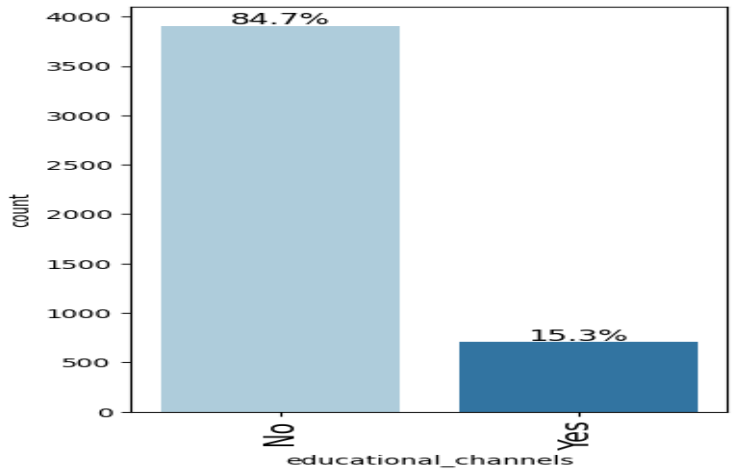

Figure 12 Observations on educational_channels

- No lead had heard about ExtraaLearn in the education channels like online forums, discussion threads, educational websites, etc. is 84.7%.
- Yes lead had heard about ExtraaLearn in the education channels like online forums, discussion threads, educational websites, etc. is 15.3%.

4.13 Observations on referral

The bar chart reflects Yes or No percentages for referral.

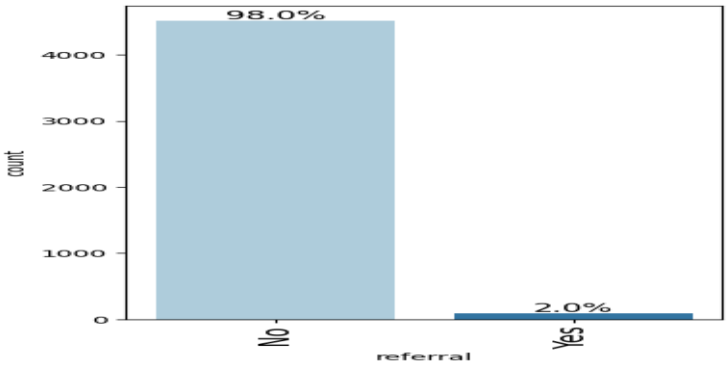

Figure 13 Observations on referral

- No lead had heard about ExtraaLearn through reference is 98.0%.
- Yes lead had heard about ExtraaLearn through reference is 2.0%.

5.0 Bivariate Analysis

Correlation Check: - we used a Heatmap to view the correlation between parameters.

Figure 14 Heatmap to view the correlation between parameters

- Time_spent_on_website and status have a strong positive correlation.
- Page_views_per_visit has a strong adverse correlation with age.
- Page_views_per_visit does not correlate with status.

5.1 Leads will have different expectations from the outcome of the course and the current occupation may play a key role for them to take the program. Let's analyze it.

The below table shows the count of status wise for current_occupation.

status	0	1	All
current_occupation			
All	3235	1377	4612
Professional	1687	929	2616
Unemployed	1058	383	1441
Student	490	65	555

The bar chart shows the status values for 0 (blue color) & 1 (orange color) respectively.

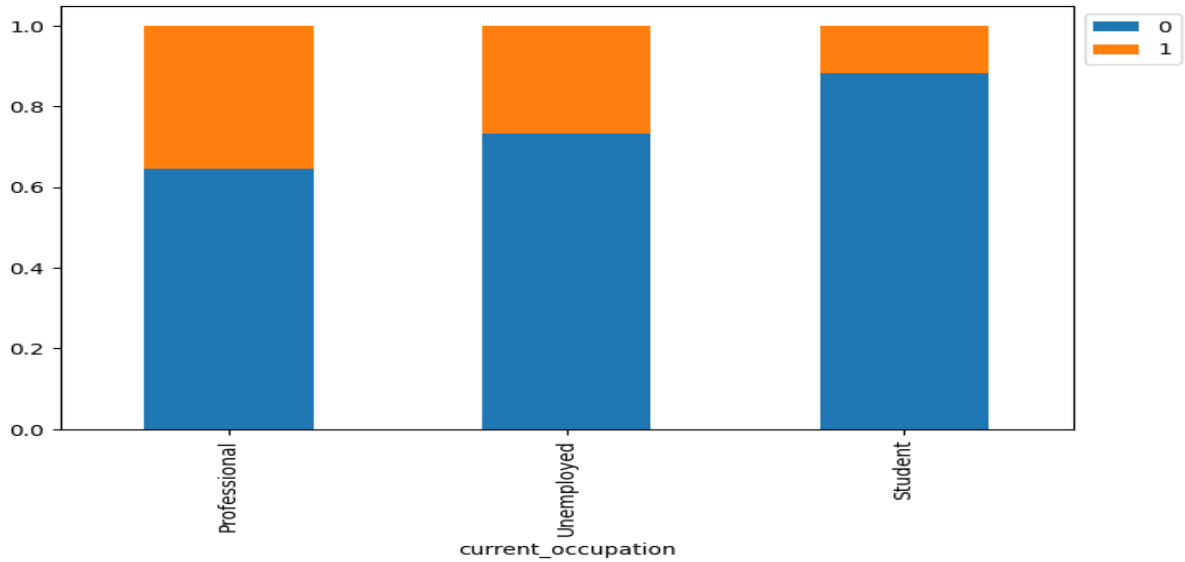

Figure 15 current_occupation & status

5.2 Boxplot between current_occupation and age

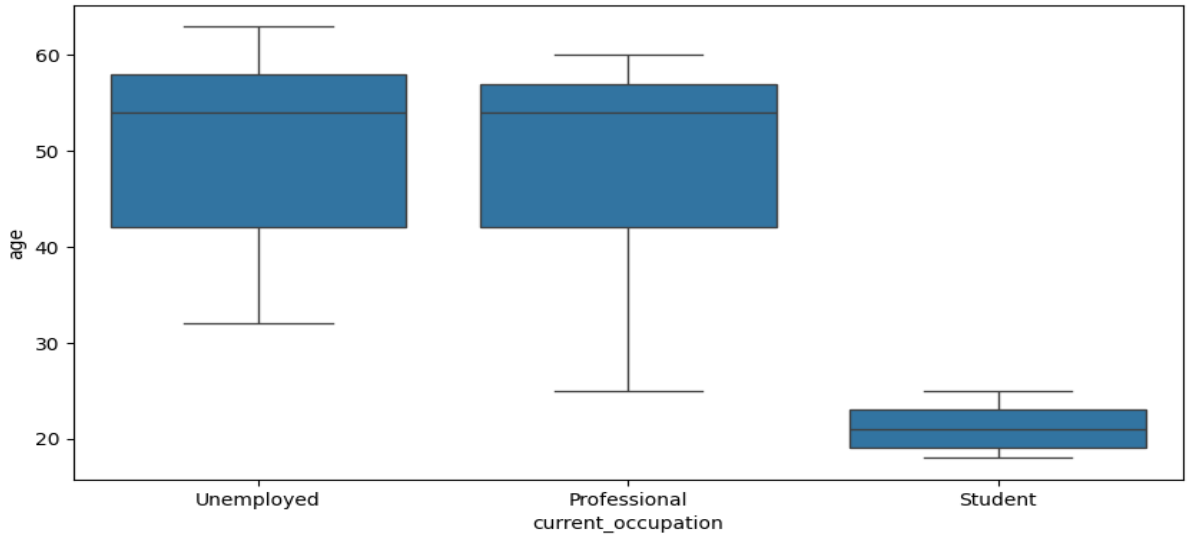

Figure 16 current_occupation and age

- There are no outliers so the data will be accurate.
- The current occupation of unemployed age is higher (57,58) than the Professional. The current occupation of student age (22) is the least.

5.3 Group by Current occupation:-

The below table is a statistical summary of the current occupation. It has grouped by professional, Student & Unemployed.

current_occupation	count	mean	std	min	25%	50%	75%	max
Professional	2616.00000	49.34748	9.89074	25.00000	42.00000	54.00000	57.00000	60.00000
Student	555.00000	21.14414	2.00111	18.00000	19.00000	21.00000	23.00000	25.00000
Unemployed	1441.00000	50.14018	9.99950	32.00000	42.00000	54.00000	58.00000	63.00000

Table 6 Group by current occupation

5.4 Time spent on website V/s Density:-

The time spent on website of the leads for target=0 has more density than target=1. Target=0 is right skewed.

Box plot for time_spent_on_website & status shows without outliers & with outliers.

Status=0 – boxplot without or with Outliers there is no change in data.

Status=1 – boxplot without & with outliers showing the same output.

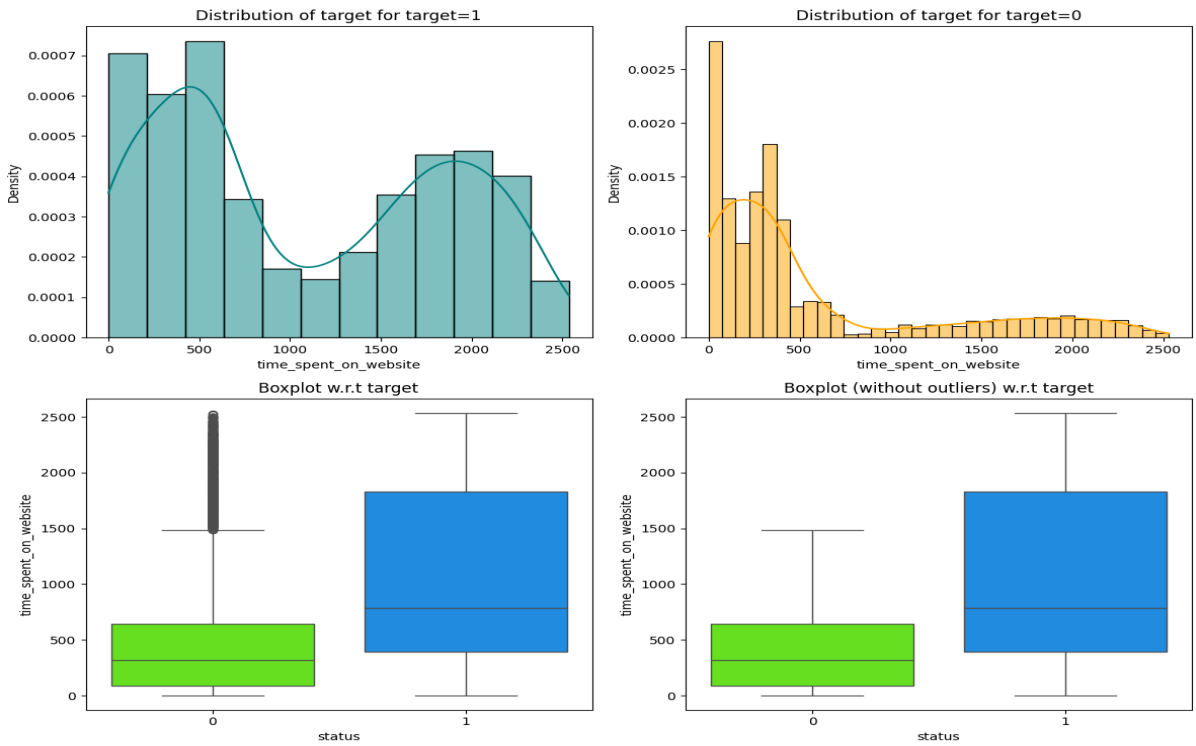

Figure 17 Time spent on website V/s Density

Overall, status-0 has outliers but status-1 has not outliers.

5.5 checking the median value:-

Time_spent_on_website for status-0 is 317 seconds.

Time_spent_on_website for status-1 is 789 seconds.

status	time_spent_on_website
0	317.00000
1	789.00000

dtype: float64

Table 7 Median value for time spent on website

5.6 To get distribution_plot_wrt_target for website_visits and status:-

The website_visits of the leads for target=0 has more density than target=1. Target=0 & Target=1 is right skewed.

Box plot for website_visits & status shows without outliers & with outliers.

Status=0 – Boxplots without & with outliers have the same number of website visits.

Status=1 – Boxplots without & with outliers have the same number of website visits.

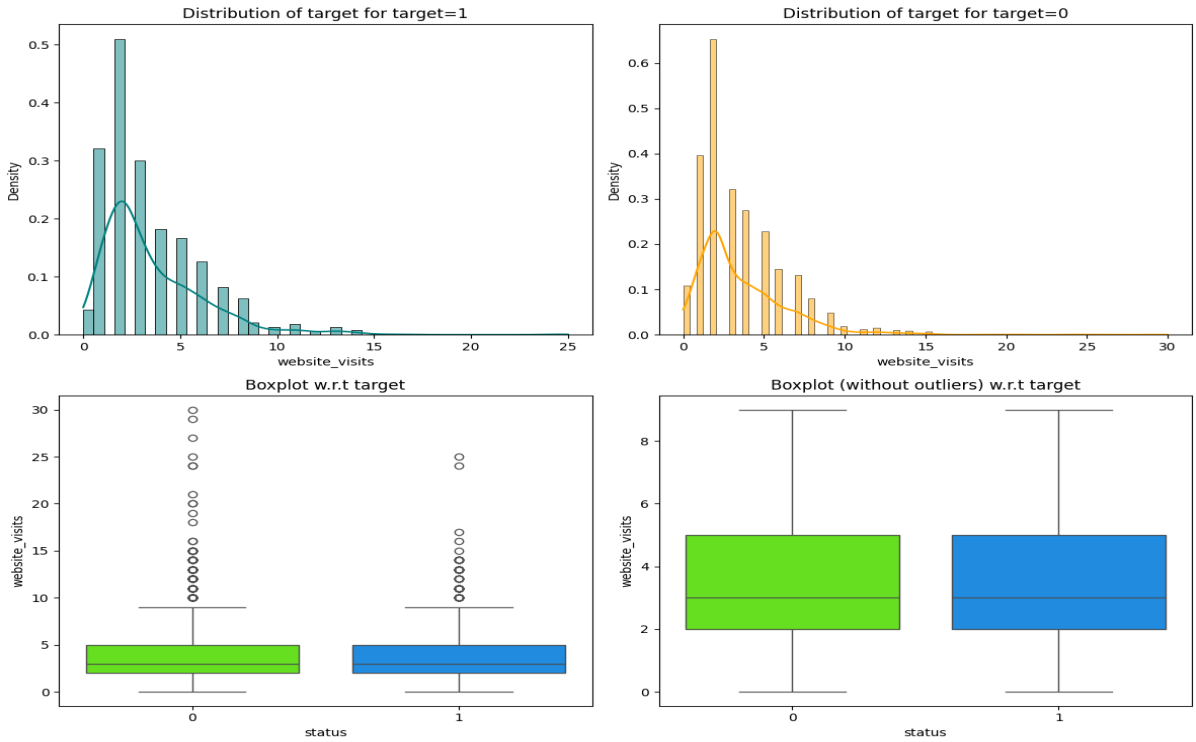

Figure 18 distribution_plot_wrt_target for website_visits and status

Overall, status-0 has more outliers than status-1.

5.7 To get distribution_plot_wrt_target for page_views_per_visit and status:-

The page_views_per_visit of the leads for target=0 has more density than target=1. Target=0 & Target=1 is right skewed.

Box plot for page_views_per_visit & status shows without outliers & with outliers.

Status=0 – Boxplots without & with outliers have the same number of page_views_per_visit.

Status=1 – Boxplots without & with outliers have the same number of page_views_per_visit.

Overall, status-0 has more outliers than status-1.

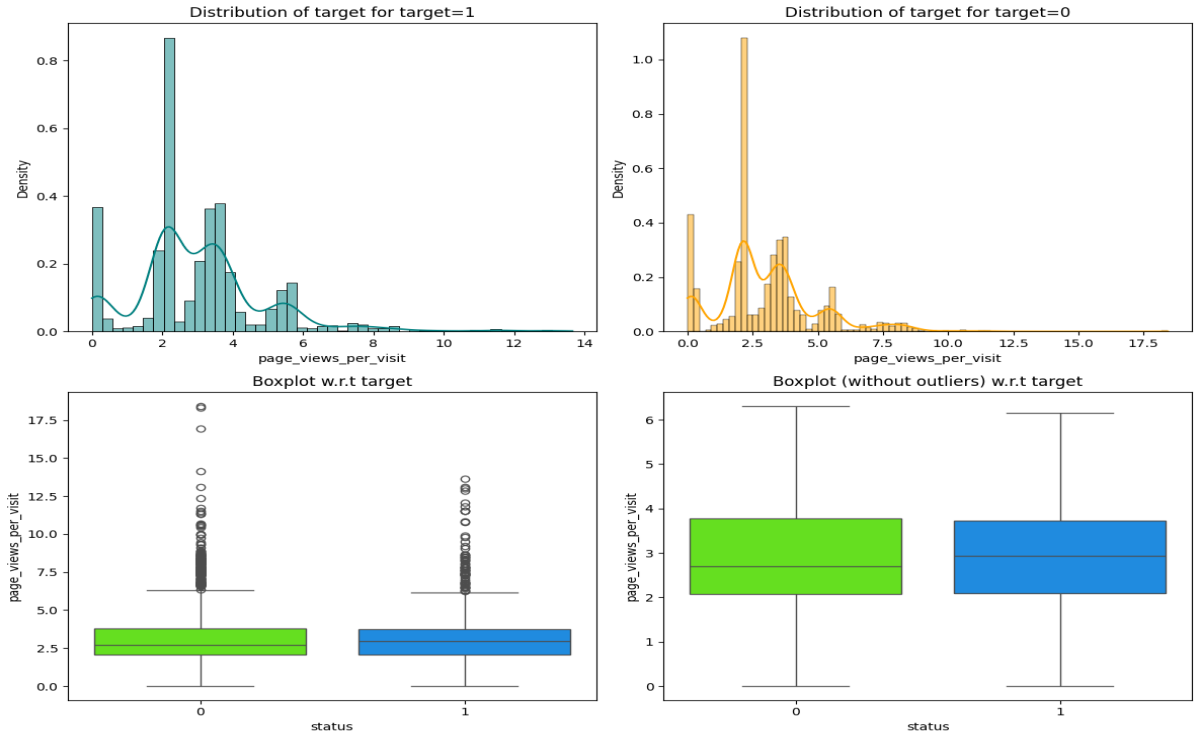

Figure 19 distribution_plot_wrt_target for page_views_per_visit and status

5.8 To get stacked_barplot for profile_completed and status

The table shows status count of profile_completed.

status	0	1	All
profile_completed			
All	3235	1377	4612
High	1318	946	2264
Medium	1818	423	2241
Low	99	8	107

Table 8 Status wise Profile completed

The conversion rate of the completed profile of status 1 is more than the medium & low.

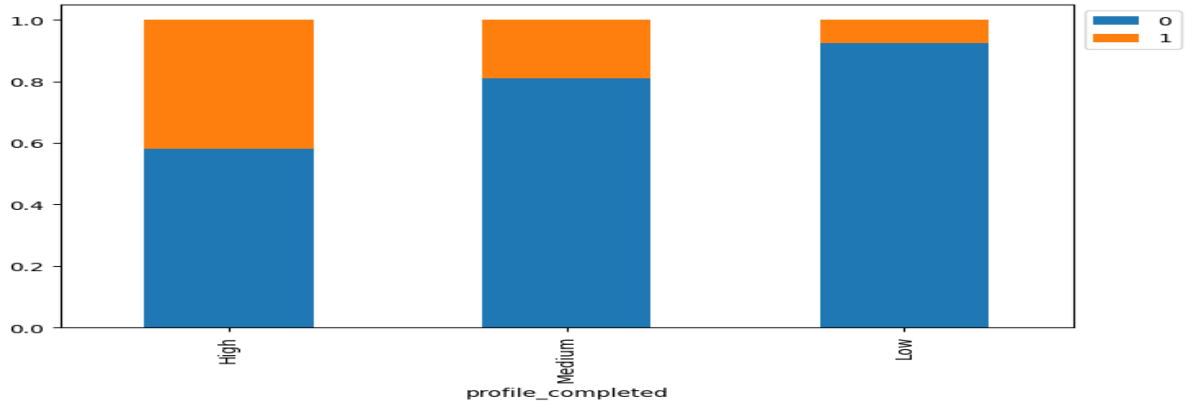

Figure 20 stacked_barplot for profile_completed and status

5.9 To get stacked_barplot for last_activity and status

The table shows status count of last_activity.

status	0	1	All
last_activity			
All	3235	1377	4612
Email Activity	1587	691	2278
Website Activity	677	423	1100
Phone Activity	971	263	1234

Table 9 status wise last activity

The conversion rate of the website_activity of status 1 is more than the email & phone activity.

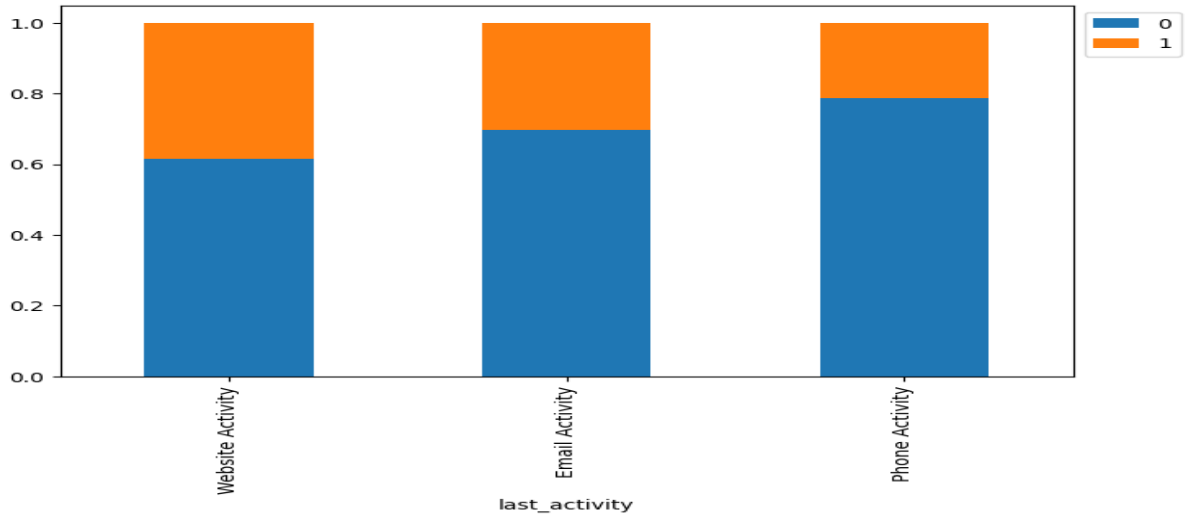

Figure 21 stacked_barplot for last_activity and status

5.10 To get stacked_barplot between print_media_type1 and status

The table shows status count of print_media_type1 (both Yes & No).

status	0	1	All
print_media_type1			
All	3235	1377	4612
No	2897	1218	4115
Yes	338	159	497

Table 10 status wise print media type1

The conversion rate of the (Yes) print_media_type1 of status 1 is more than the others (No).

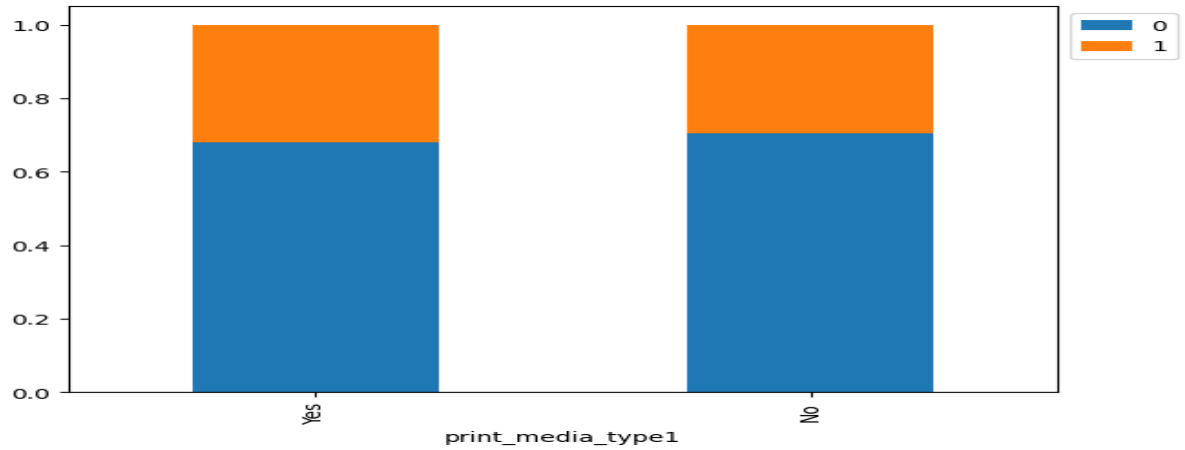

Figure 22 stacked_barplot between print_media_type1 and status

5.11 To get stacked_barplot between print_media_type1 and status

The table shows status count of print_media_type1 (both Yes & No).

status	0	1	All
print_media_type2			
All	3235	1377	4612
No	3077	1302	4379
Yes	158	75	233

Table 11 status wise print media type2

The conversion of the Yes - print_media_type2 of status 1 is more than the others (No).

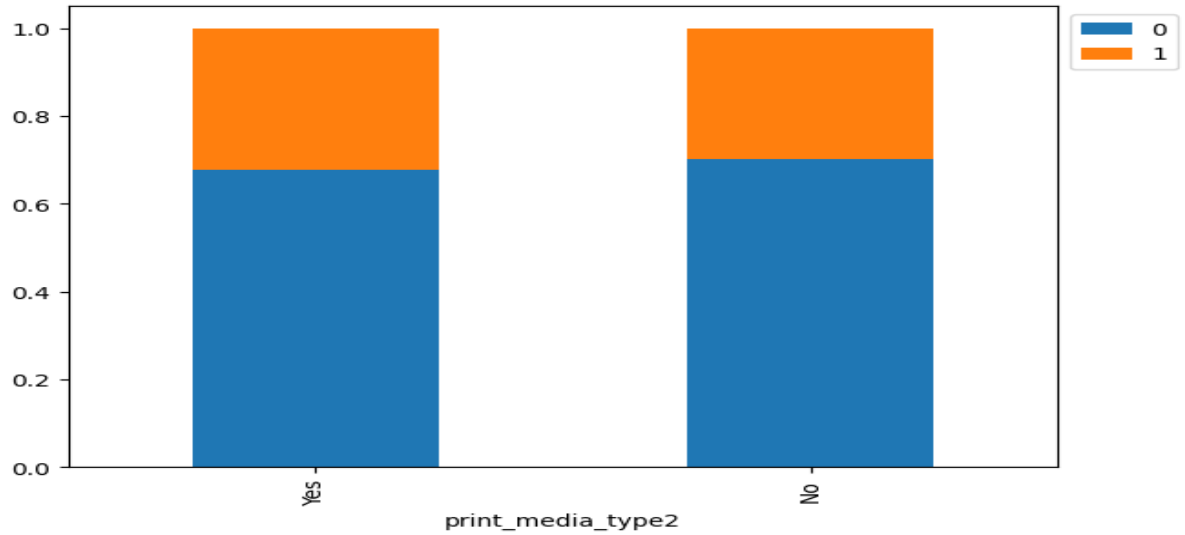

Figure 23 stacked_barplot between print_media_type1 and status

5.12 To get stacked_barplot between digital_media and status:-

The table shows status count of print_media_type1 (both Yes & No).

status	0	1	All
digital_media			
All	3235	1377	4612
No	2876	1209	4085
Yes	359	168	527

Table 12 status wise digital media count

The conversion of the Yes – digital_media of status 1 is more than the others (No).

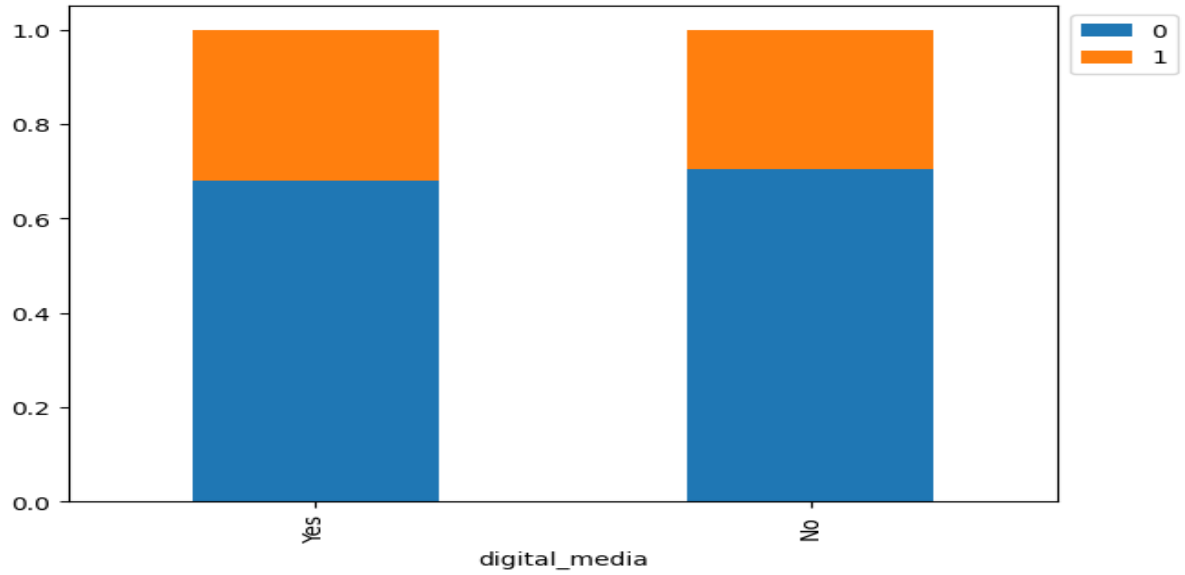

Figure 24 stacked_barplot between digital_media and status

5.13 To get stacked_barplot between educational_channels and status

The table shows status count of educational_channels (both Yes & No).

status	0	1	All
educational_channels			
All	3235	1377	4612
No	2727	1180	3907
Yes	508	197	705

Table 13 status wise educational channel count

The conversion of the educational_channels - Yes of status 1 is more than the others (No).

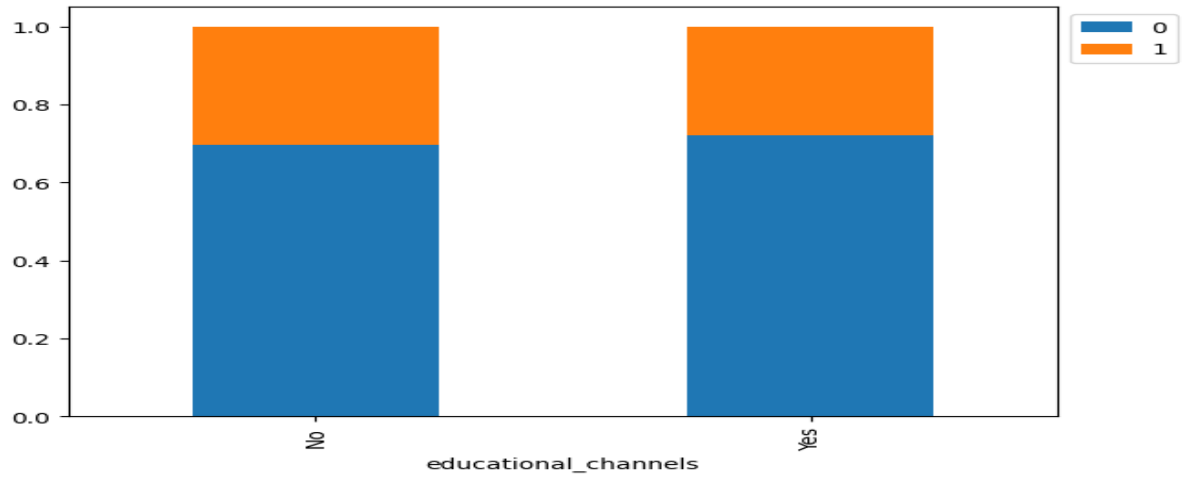

Figure 25 stacked_barplot between educational_channels and status

5.14 To get stacked_barplot between referral and status

The table shows status count of referral (both Yes & No).

status	0	1	All
referral			
All	3235	1377	4612
No	3205	1314	4519
Yes	30	63	93

Table 14 status wise count for referral

The conversion of the referral - Yes of status 1 is more than the others (No).

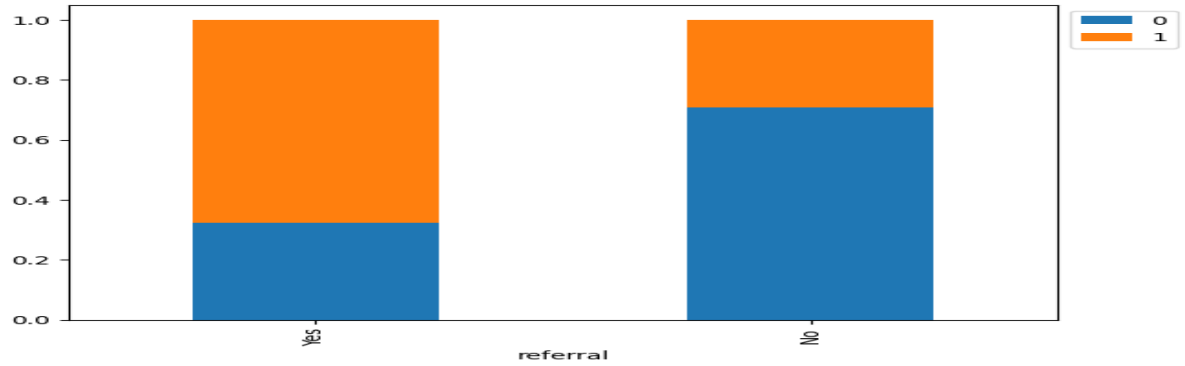

Figure 26 stacked_barplot between referral and status

6.0 Data Preprocessing

6.1 Outlier Check:- Let's check for outliers in the data.

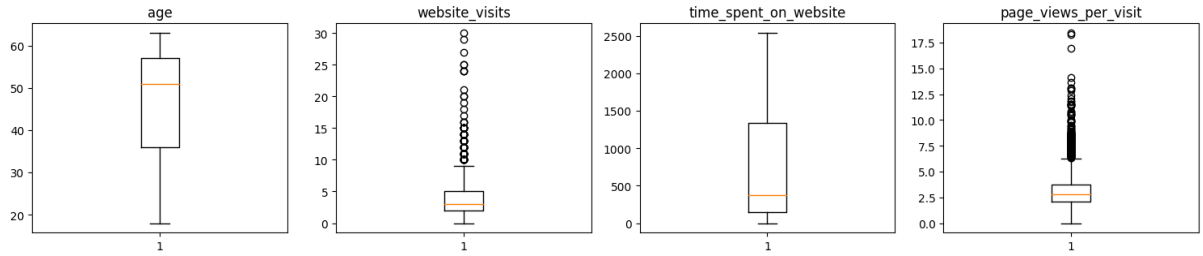

Figure 27 Outlier Check

website_visits & page_views_per_visit have many outliers so not preferable. However, there are no outliers for age & time_spent_on_website but can be preferable.

6.2 Data Preparation for modeling

- We want to predict which lead is more likely to be converted.
- Before we proceed to build a model, we'll have to encode categorical features.
- We'll split the data into train and test to be able to evaluate the model that we build on the train data.

To split the data into train and test in the ratio 75:25.

```
Shape of Training set :  (3459, 16)
Shape of test set :  (1153, 16)
Shape of Training set :  (3459,)
Shape of test set :  (1153,)
Percentage of classes in training set:
status
0   0.70107
1   0.29893
Name: proportion, dtype: float64
Percentage of classes in test set:
status
0   0.70252
1   0.29748
Name: proportion, dtype: float64
```

Table 15 split data into train & test ratio 75:25

Scaling the Data – It is standard scaler.

7.0 Model Building

Model evaluation criterion

Evaluation Metric of Choice

- First, let's create functions to calculate different metrics and confusion matrix so that we don't have to use the same code repeatedly for each model.
- The model_performance_classification function will be used to check the model performance of models.
- The plot_confusion_matrix function will be used to plot the confusion matrix.

7.1 Logistic Regression (with Statsmodel)

```
Optimization terminated successfully.
         Current function value: 0.393259
         Iterations 7
                           Logit Regression Results
==============================================================================
Dep. Variable:                 status   No. Observations:                 3459
Model:                          Logit   Df Residuals:                     3442
Method:                           MLE   Df Model:                           16
Date:                Fri, 11 Oct 2024   Pseudo R-squ.:                  0.3553
Time:                        06:22:49   Log-Likelihood:                -1360.3
converged:                       True   LL-Null:                       -2109.8
Covariance Type:            nonrobust   LLR p-value:                 7.912e-310
==========================================================================================
                                     coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                             -1.4531      0.059    -24.504      0.000      -1.569      -1.337
age                                0.0734      0.067      1.103      0.270      -0.057       0.204
website_visits                    -0.0226      0.050     -0.457      0.648      -0.120       0.075
time_spent_on_website              0.9652      0.051     18.907      0.000       0.865       1.065
page_views_per_visit              -0.0493      0.050     -0.988      0.323      -0.147       0.048
current_occupation_Student        -0.6201      0.079     -7.887      0.000      -0.774      -0.466
current_occupation_Unemployed     -0.2473      0.049     -5.029      0.000      -0.344      -0.151
first_interaction_Website          1.3399      0.060     22.306      0.000       1.222       1.458
profile_completed_Low             -0.3942      0.076     -5.214      0.000      -0.542      -0.246
profile_completed_Medium          -0.8002      0.052    -15.472      0.000      -0.902      -0.699
last_activity_Phone Activity      -0.3013      0.054     -5.608      0.000      -0.407      -0.196
last_activity_Website Activity     0.2213      0.049      4.492      0.000       0.125       0.318
print_media_type1_Yes              0.0615      0.046      1.331      0.183      -0.029       0.152
print_media_type2_Yes              0.0433      0.046      0.942      0.346      -0.047       0.134
digital_media_Yes                  0.0253      0.047      0.542      0.588      -0.066       0.117
educational_channels_Yes           0.0287      0.049      0.591      0.555      -0.067       0.124
referral_Yes                       0.1868      0.048      3.879      0.000       0.092       0.281
==========================================================================================
```

Table 16 Logistic Regression Model Building

The P-value greater than 0.05 will drop from the data are digital_media_Yes - 0.0588, educational_channels_Yes – 0.0555.

7.1.1 Checking Logistic Regression model performance on training set

	Accuracy	Recall	Precision	F1
0	0.82249	0.65377	0.72532	0.68769

Table 17 Logistic regression model performance training set

The recall is 65.37% so it is not much better to proceed for training set.

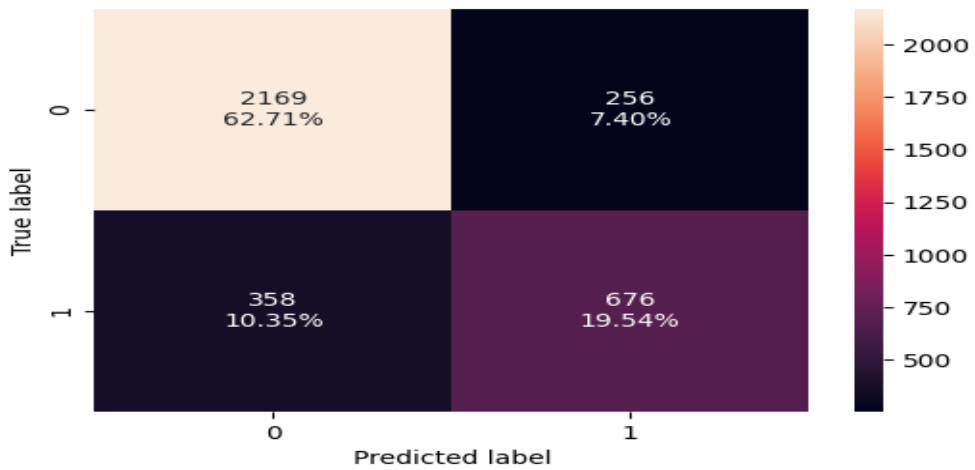

Figure 28 confusion_matrix training data logistic regression

True Negative is – 62.71% which is high so it is not a good percentage for training data.

7.1.2 Checking Logistic Regression model performance on test set.

The recall is 61.22% so it is not much better to proceed for training set.

	Accuracy	Recall	Precision	F1
0	0.80399	0.61224	0.69307	0.65015

Table 18 Logistic regression model performance test set

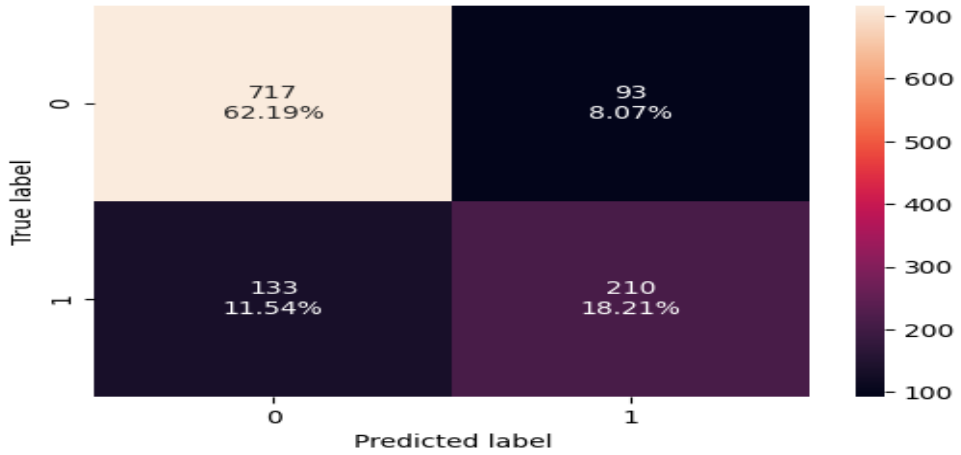

Figure 29 Logistic Regression model performance on test set

True Negative is – 62.19% which is high so it is not a good percentage for test data.

7.2 Naive- Bayes Classifier

7.2.1 Build Naive Bayes Model

```
GaussianNB
GaussianNB()
```

7.2.2 Checking Naive- Bayes Classifier performance on training set

To get model performance on training data

The recall is 76.88% so it is not much better to proceed for training set.

	Accuracy	Recall	Precision	F1
0	0.78925	0.76886	0.61868	0.68564

Table 19 Naive - Bayes Classifier performance on training set

To create a confusion matrix for training data

The recall is 55.94% so it is not much better to proceed for training set.

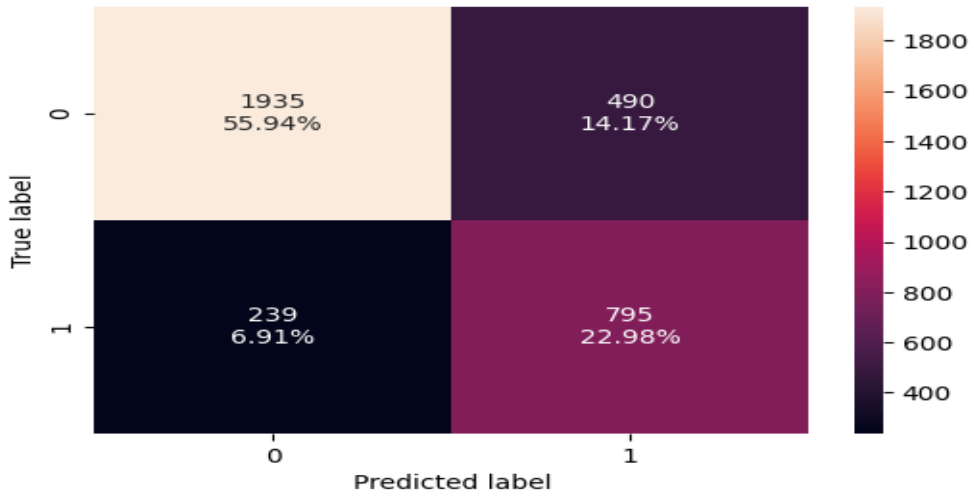

Figure 30 Naive - Bayes Classifier performance on training set

7.2.3 Checking Naive- Bayes Classifier performance on test set

To get model performance on test data

The recall is 77.97% so it is not much better to proceed for test set which is near to training set.

	Accuracy	Recall	Precision	F1
0	0.77971	0.75802	0.60325	0.67183

Table 20 Naive - Bayes Classifier performance on test set

To create confusion matrix for test data

The True Negative is 55.42% so it is not much better to proceed for test set and no much difference for training set.

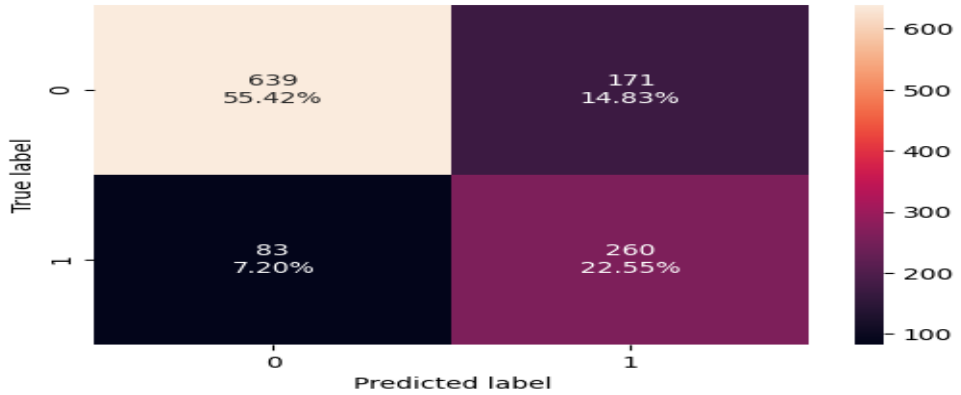

Figure 31 Naive - Bayes Classifier performance on training set confusion matrix

7.3 KNN Classifier (K = 3)

To build KNN model with nummber of neighbors as 3

```
KNeighborsClassifier
KNeighborsClassifier(n_neighbors=3)
```

7.3.1 Checking KNN Classifier performance on training set

The recall is 82.23% so it is better to proceed for training set but not better to depend on it completely.

	Accuracy	Recall	Precision	F1
0	0.47355	0.81238	0.34049	0.47986

Table 21 KNN Classifier performance on training set

To create a confusion matrix for training data

The true negative is 23.07% so it is not much better to proceed with test set and the accuracy will be less.

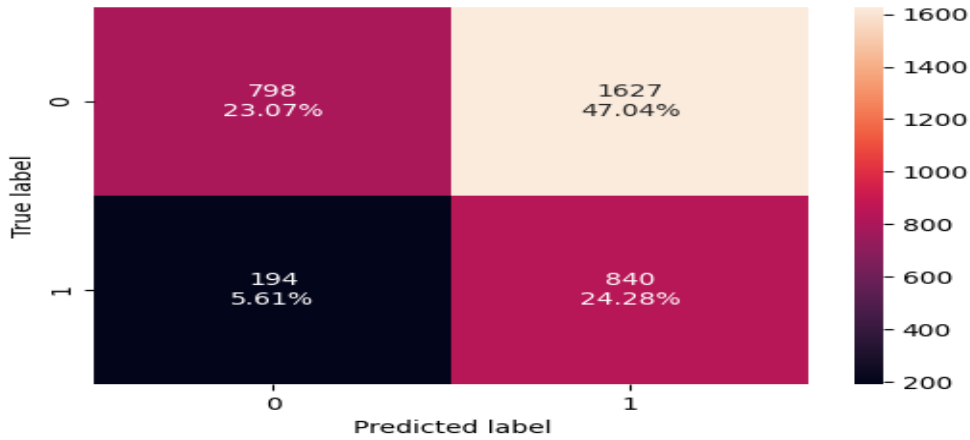

Figure 32 KNN Classifier (K = 3) confusion matrix training data

7.3.2 Checking KNN Classifier performance on test set

To get model performance on test data

The recall is 79.00 % so it is not much better to proceed for test set which has less accuracy.

	Accuracy	Recall	Precision	F1
0	0.46748	0.79009	0.33333	0.46886

Table 22 KNN Classifier performance on test set

To create confusion matrix for test data

The True Negative is 23.24% so it is not much better to proceed for the test set and no much difference with training set.

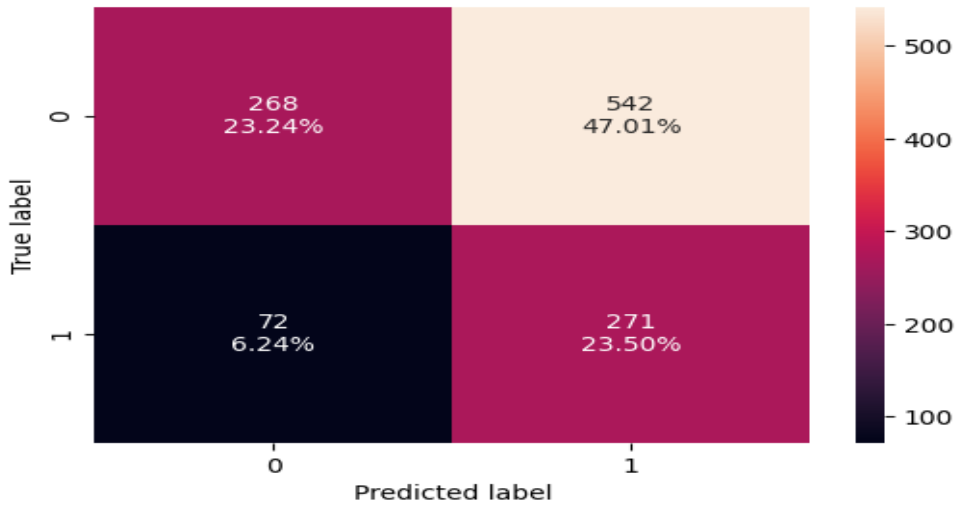

Figure 33 KNN Classifier (K = 3) confusion matrix test data

7.4 Decision Tree Classifier

```
DecisionTreeClassifier
DecisionTreeClassifier(random_state=42)
```

7.4.1 Checking Decision Tree Classifier performance on training set

To get model performance on training data

The recall is 99.90% so it is much better to proceed for training set which has good accuracy.

	Accuracy	Recall	Precision	F1
0	0.99971	0.99903	1.00000	0.99952

Table 23 Decision Tree Classifier performance on training set

To create confusion matrix for training data

The True Negative is 70.11% so it is not much better to proceed for test set.

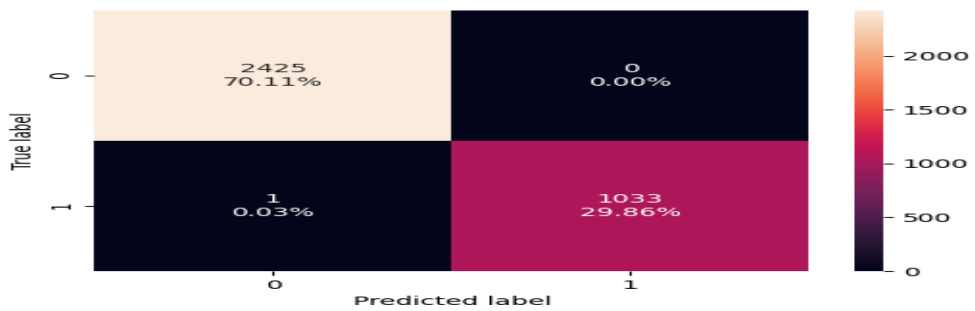

Figure 34 Decision Tree Classifier performance on training set confusion matrix

Checking Decision Tree Classifier performance on test set

7.4.2 To get model performance on test data

The recall is 67.63% so it is not much better to proceed for test set which has less accuracy.

	Accuracy	Recall	Precision	F1
0	0.81526	0.67638	0.69461	0.68538

Table 24 Decision Tree Classifier performance on test set

To create confusion matrix for test data

The True Negative is 61.41% so it is not much better to proceed for test set.

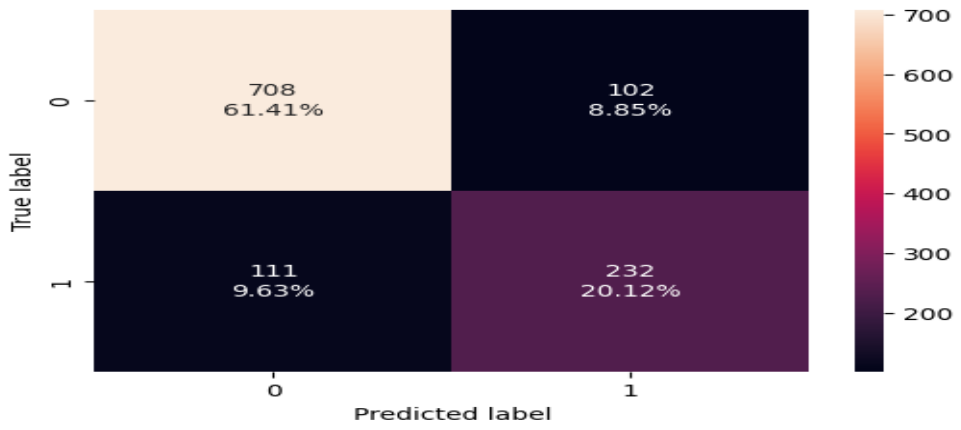

Figure 35 Decision Tree Classifier performance on test set confusion matrix

8.0 Model Performance Improvement

Logistic Regression (deal with multicollinearity, remove high p-value variables, determine optimal threshold using ROC curve)

Logistic Regression - Dealing with Multicollinearity

calculate VIF

```
Variance Inflation Factors:
                          Variable     VIF
0                              age 6.96678
1                    website_visits 2.52701
2             time_spent_on_website 1.96142
3              page_views_per_visit 3.32519
4        current_occupation_Student 1.34553
5     current_occupation_Unemployed 1.55809
6         first_interaction_Website 2.14406
7             profile_completed_Low 1.05209
8          profile_completed_Medium 1.92465
9      last_activity_Phone Activity 1.51374
10  last_activity_Website Activity 1.46403
11            print_media_type1_Yes 1.12686
12            print_media_type2_Yes 1.05689
13                digital_media_Yes 1.12899
14         educational_channels_Yes 1.18695
15                     referral_Yes 1.03004
```

Table 25 Calculate VIF

Note: If you find any variables with high VIF and wish to drop them, please uncomment and run the below code for the same. Feel free to modify the below code according to your own requirements.

Dealing with high p-value variables

```
Optimization terminated successfully.
         Current function value: 0.393259
         Iterations 7
Dropping column website_visits with p-value: 0.6479479510695756
Optimization terminated successfully.
         Current function value: 0.393289
         Iterations 7
Dropping column digital_media_Yes with p-value: 0.5827477567779233
Optimization terminated successfully.
         Current function value: 0.393332
         Iterations 7
Dropping column educational_channels_Yes with p-value: 0.5567480874402153
Optimization terminated successfully.
         Current function value: 0.393382
         Iterations 7
Dropping column print_media_type2_Yes with p-value: 0.33884314243659996
Optimization terminated successfully.
         Current function value: 0.393513
         Iterations 7
Dropping column page_views_per_visit with p-value: 0.32588281893475446
Optimization terminated successfully.
         Current function value: 0.393653
         Iterations 7
Dropping column age with p-value: 0.28978634672139825
Optimization terminated successfully.
         Current function value: 0.393816
         Iterations 7
Dropping column print_media_type1_Yes with p-value: 0.1535692252215647
Optimization terminated successfully.
         Current function value: 0.394108
         Iterations 7
Dropping column referral_Yes with p-value: 8.577120192655911e-05
['const', 'time_spent_on_website', 'current_occupation_Student', 'current_occupation_Unemployed', 'first_interaction_Website', 'profile_completed_Low', 'profile_completed_Medium', 'last_activity_Phone Activity', 'last_activity_Website Activity', 'referral_Yes']
```

Table 26 High p-value variables

To get the test data with selected features

	const	time_spent_on_website	current_occupation_Student	current_occupation_Unemployed	first_interaction_Website	profile_completed_Low	profile_completed_Medium	last_activity_Phone Activity	last_activity_Website Activity	referral_Yes
0	1.00000	0.74345	-0.36873	1.46443	-1.09882	-0.15776	-0.96561	1.66496	-0.56322	-0.14267
1	1.00000	-0.97204	2.71202	-0.68286	0.91007	-0.15776	1.03562	-0.60062	-0.56322	-0.14267
2	1.00000	-0.57327	-0.36873	-0.68286	-1.09882	-0.15776	-0.96561	-0.60062	-0.56322	-0.14267
3	1.00000	-0.20930	-0.36873	-0.68286	-1.09882	-0.15776	1.03562	-0.60062	1.77552	-0.14267
4	1.00000	-0.97204	-0.36873	1.46443	0.91007	-0.15776	1.03562	1.66496	-0.56322	-0.14267
5	1.00000	-0.97070	-0.36873	-0.68286	0.91007	-0.15776	-0.96561	-0.60062	-0.56322	-0.14267
6	1.00000	1.59316	-0.36873	-0.68286	-1.09882	-0.15776	-0.96561	1.66496	-0.56322	-0.14267
7	1.00000	0.20686	-0.36873	-0.68286	-1.09882	-0.15776	1.03562	-0.60062	-0.56322	-0.14267
8	1.00000	0.87726	-0.36873	1.46443	0.91007	-0.15776	-0.96561	1.66496	-0.56322	-0.14267
9	1.00000	0.41560	-0.36873	-0.68286	-1.09882	-0.15776	1.03562	-0.60062	-0.56322	-0.14267

Table 27 test data with selected features

9.0 Training the Logistic Regression model again with only the significant features

To train the Logistic Regression model with significant features

```
Optimization terminated successfully.
         Current function value: 0.394108
         Iterations 7
                           Logit Regression Results
==============================================================================
Dep. Variable:                 status   No. Observations:                 3459
Model:                          Logit   Df Residuals:                     3449
Method:                           MLE   Df Model:                            9
Date:                Fri, 11 Oct 2024   Pseudo R-squ.:                  0.3539
Time:                        06:28:48   Log-Likelihood:                -1363.2
converged:                       True   LL-Null:                       -2109.8
Covariance Type:            nonrobust   LLR p-value:                     0.000
==========================================================================================
                                     coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                             -1.4470      0.059    -24.525      0.000      -1.563      -1.331
time_spent_on_website              0.9609      0.051     18.961      0.000       0.862       1.060
current_occupation_Student        -0.6742      0.063    -10.728      0.000      -0.797      -0.551
current_occupation_Unemployed     -0.2458      0.049     -5.019      0.000      -0.342      -0.150
first_interaction_Website          1.3373      0.060     22.329      0.000       1.220       1.455
profile_completed_Low             -0.3958      0.075     -5.254      0.000      -0.543      -0.248
profile_completed_Medium          -0.8001      0.052    -15.496      0.000      -0.901      -0.699
last_activity_Phone Activity      -0.3022      0.054     -5.639      0.000      -0.407      -0.197
last_activity_Website Activity     0.2186      0.049      4.456      0.000       0.122       0.315
referral_Yes                       0.1889      0.048      3.928      0.000       0.095       0.283
==========================================================================================
```

Table 28 Logistic Regression model again with only the significant features

It is a good model because there is no P-Value greater than 0.05.

9.1 Determining optimal threshold using ROC Curve

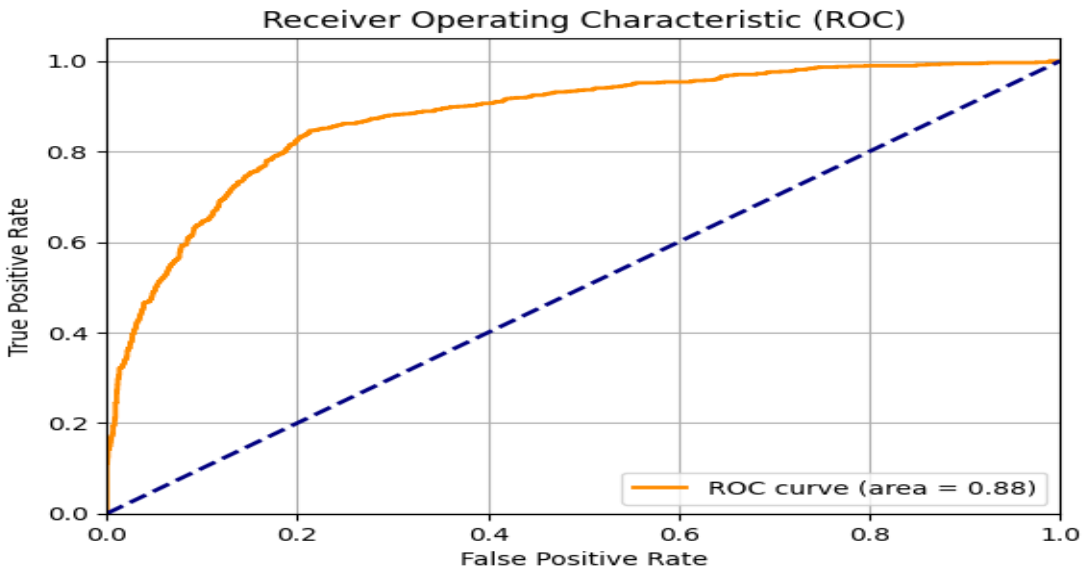

Figure 36 optimal threshold using ROC Curve

9.3 Find the optimal threshold

Optimal Threshold: 0.282

Checking tuned Logistic Regression model performance on training set

The recall is 84.62% so it is better to proceed for training set but the accuracy is less.

	Accuracy	Recall	Precision	F1
0	0.80457	0.84623	0.62859	0.72135

Table 29 tuned Logistic Regression model performance on training set

The True Negative is 55.16% so it is not much better to proceed for training set.

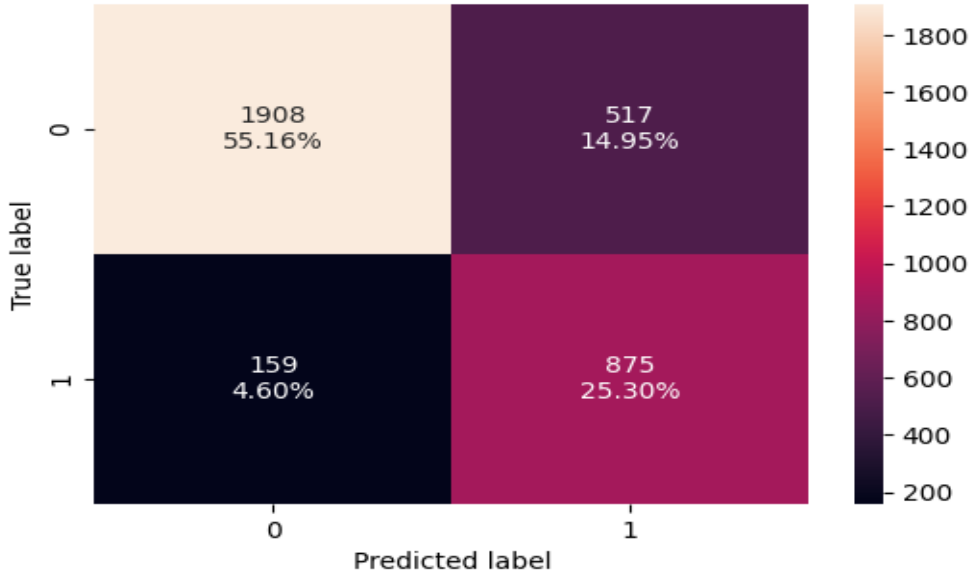

Figure 37 Logistic Regression model performance on training set confusion matrix

9.4 Checking tuned Logistic Regression model performance on test set

The recall is 83.09% so it is better to proceed for test set but the accuracy is less.

	Accuracy	Recall	Precision	F1
0	0.79098	0.83090	0.60897	0.70284

Table 30 tuned Logistic Regression model performance on test set

The True Negative is 54.38% so it is not much better to proceed for test set.

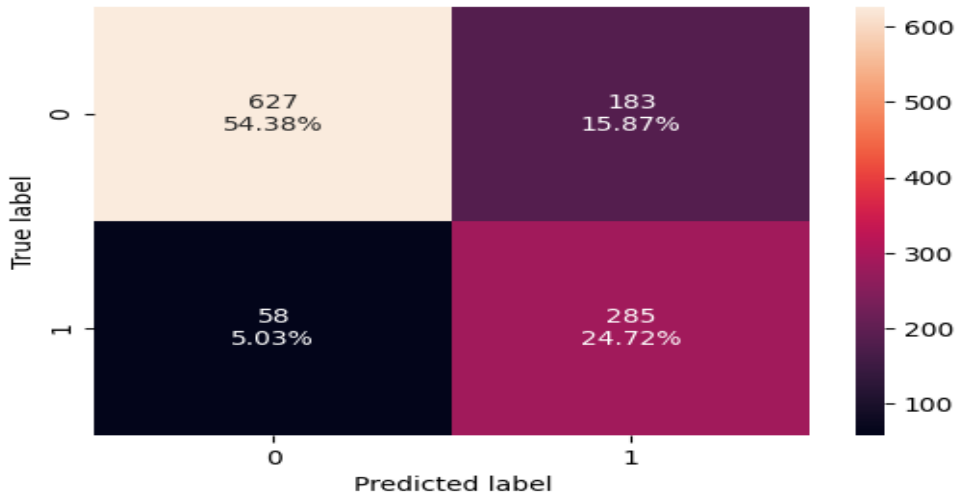

Figure 38 tuned Logistic Regression model performance on test set confusion matrix

10.0 KNN Classifier (different values of K)

KNN Classifier Performance Improvement using different k values

to define the range for k-values between 2 and 20 (both inclusive)

The best k value and its recall score

```
Recall for k=2: 0.43440233236151604
Recall for k=3: 0.597667638483965
Recall for k=4: 0.5043731778425656
Recall for k=5: 0.6239067055393586
Recall for k=6: 0.5539358600583091
Recall for k=7: 0.6530612244897959
Recall for k=8: 0.5918367346938775
Recall for k=9: 0.6588921282798834
Recall for k=10: 0.5743440233236151
Recall for k=11: 0.641399416909621
Recall for k=12: 0.5685131195335277
Recall for k=13: 0.6209912536443148
Recall for k=14: 0.565597667638484
Recall for k=15: 0.6064139941690962
Recall for k=16: 0.5714285714285714
Recall for k=17: 0.6297376093294461
Recall for k=18: 0.5860058309037901
Recall for k=19: 0.6180758017492711
Recall for k=20: 0.597667638483965

The best value of k is: 9 with a recall of: 0.6588921282798834
```

Table 31 best k value and its recall score

10.1 To build KNN model with number of neighbors as best_k

To build KNN model with number of neighbors as best k

```
KNeighborsClassifier
KNeighborsClassifier(n_neighbors=9)
```

10.2 Checking tuned KNN model performance on training set

The recall is 68.18% so it is not better to proceed for training set but the accuracy is more so it will be useful.

	Accuracy	Recall	Precision	F1
0	0.85256	0.68182	0.79571	0.73438

Table 32 tuned KNN model performance on training set

The True Negative is 4.77% so it is not much better to proceed for training set.

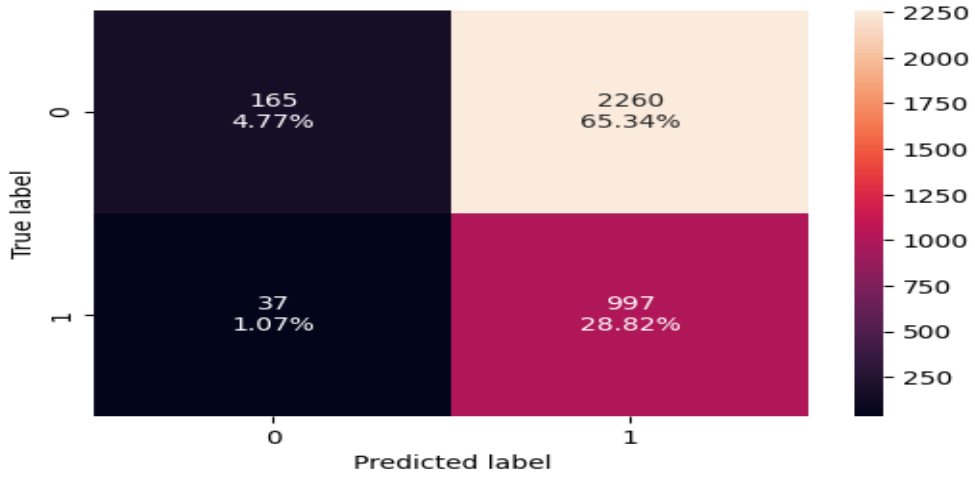

Figure 39 tuned KNN model performance on training set confusion matrix

10.3 Checking tuned KNN model performance on test set

The recall is 65.88% so it is not better to proceed for test set but the accuracy is more.

	Accuracy	Recall	Precision	F1
0	0.83695	0.65889	0.76094	0.70625

Table 33 tuned KNN model performance on test set

The True Negative is 61.41% so it is not much better to proceed for test set.

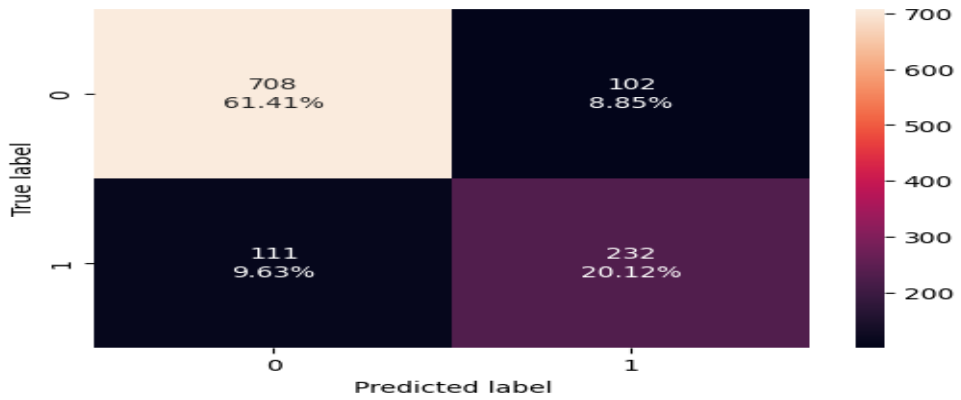

Figure 40 tuned KNN model performance on test set confusion matrix

11.0 Decision Tree Classifier (pre-pruning)

Pre-pruning the tree

Note: The parameter values provided are a sample. You can feel free to add/remove values to experiment with different parameter combinations.

```
DecisionTreeClassifier
DecisionTreeClassifier(class_weight='balanced', max_depth=5, max_leaf_nodes=40,
                       min_samples_split=20, random_state=42)
```

11.1 Checking tuned Decision Tree Classifier performance on training set

Recall and Accuracy are both near each other so it can proceed.

	Accuracy	Recall	Precision	F1
0	0.83463	0.88685	0.66837	0.76226

Table 34 tuned Decision Tree Classifier performance on training set

To create confusion matrix for training data

It has True negative more than others so it not preferable.

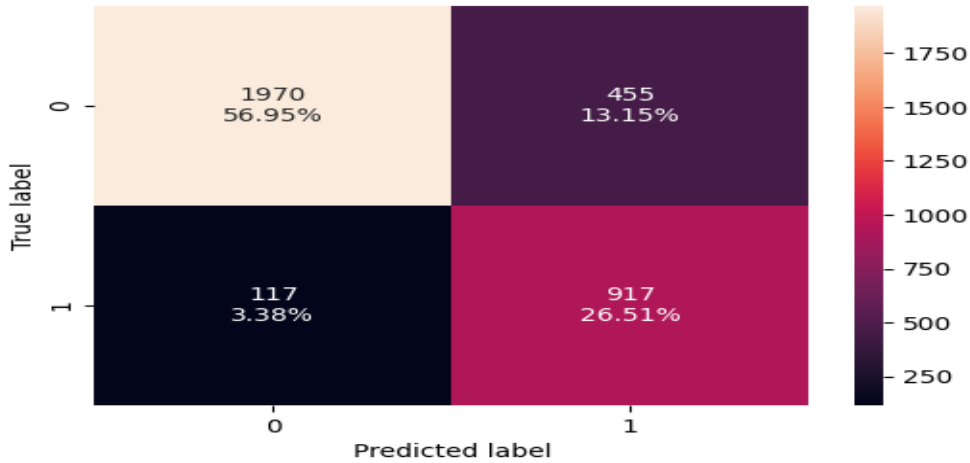

Figure 41 tuned Decision Tree Classifier performance on training set confusion matrix

11.2 Checking tuned Decision Tree Classifier performance on test set

Recall is more but the accuracy is also good so it can be preferable.

	Accuracy	Recall	Precision	F1
0	0.83955	0.89213	0.67401	0.76788

Table 35 tuned Decision Tree Classifier performance on test set

To create confusion matrix for test data

It has True negative more than others so it not preferable.

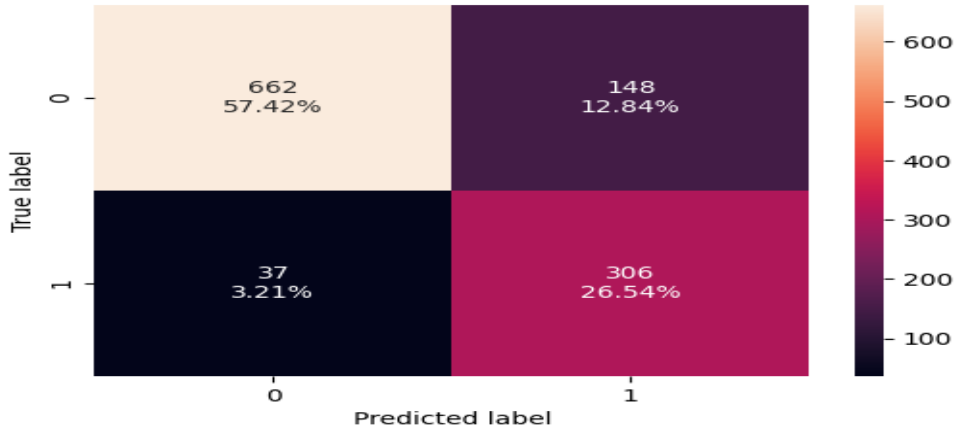

Figure 42 tuned Decision Tree Classifier performance on test set confusion matrix

12.0 Visualizing the Decision Tree

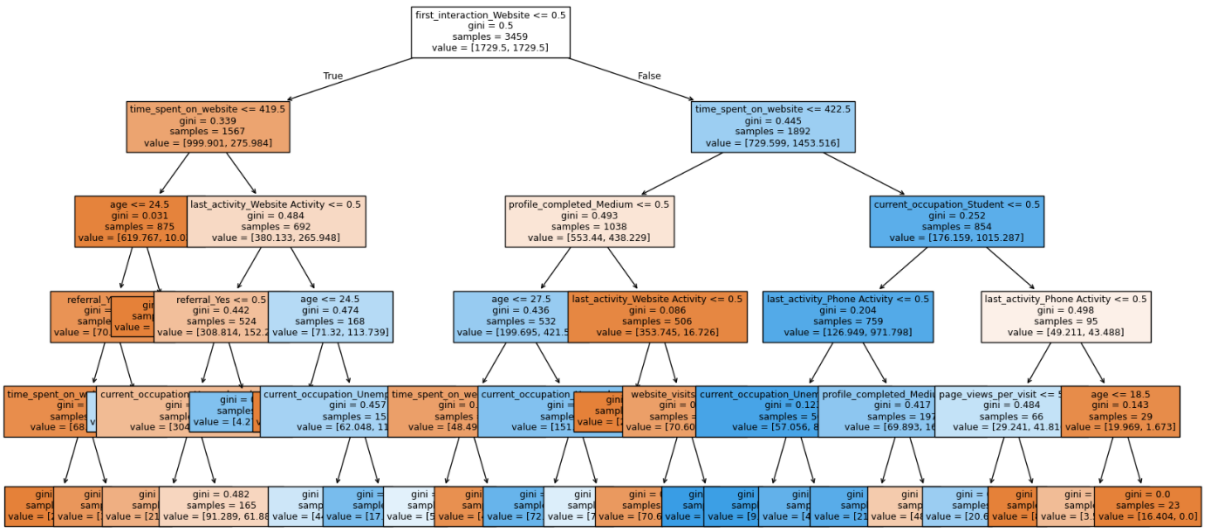

Figure 43 Visualizing the Decision Tree

Analyzing Feature Importance for tuned Decision Tree Classifier

The tuned Decision Tree Classifier is more complex due too many branches, leaf's.

13.0 Model Performance Comparison and Final Model Selection

Training performance comparison:

	Logistic Regression Base	Logistic Regression Tuned	Naive Bayes Base	KNN Base	KNN Tuned	Decision Tree Base	Decision Tree Tuned
Accuracy	0.82249	0.80457	0.78925	0.47355	0.85256	0.99971	0.83463
Recall	0.65377	0.84623	0.76886	0.81238	0.68182	0.99903	0.88685
Precision	0.72532	0.62859	0.61868	0.34049	0.79571	1.00000	0.66837
F1	0.68769	0.72135	0.68564	0.47986	0.73438	0.99952	0.76226

Table 36 Training performance comparison:

Test set performance comparison:

	Logistic Regression Base	Logistic Regression Tuned	Naive Bayes Base	KNN Base	KNN Tuned	Decision Tree Base	Decision Tree Tuned
Accuracy	0.80399	0.79098	0.77971	0.46748	0.83695	0.99971	0.83955
Recall	0.61224	0.83090	0.75802	0.79009	0.65889	0.99903	0.89213
Precision	0.69307	0.60897	0.60325	0.33333	0.76094	1.00000	0.67401
F1	0.65015	0.70284	0.67183	0.46886	0.70625	0.99952	0.76788

Table 37 Test set performance comparison

14.0 Actionable Insights and Recommendations

14.1 Actionable Insights:-

- Outliers are more on website visits & page views per visit so it is not better to prefer.
- P-value is higher than 0.05 in the logistic regression model which reduces the performance.
- The confusion matrix for training data for True Negative is high in many cases which makes the model complex.
- In many cases, Recall the model performance on test data is lower than accuracy.
- Visualizing the Decision tree is complex for the post-pruning the tree.
- ML 1 Models need to be built the model repeatedly due to dissatisfaction with the rules.
- Targeting the specific age group.
- Less conversion rate for the profile completed status.
- As per the observations on the last activity email activity is more than phone & website activity.
- The number of page views per visit is more flexible and gradually reducing.
- No status for print media is high for both newspaper & magazine.

14.2 Recommendations:-

- Mostly consider the variables that do not have outliers for the effectiveness & accuracy of the data.
- Model performance is good wherever the P-value is less than 0.05.

- It is not better to follow if the True negative is high.
- Accuracy model performance on training data is higher than Recall, it is better to proceed.
- It is better to use Pre-pruning the tree which is very easy for decision making.
- ML 1 Models are required to satisfy the rules immediately to make it effective and to avoid.
- Spread awareness to all age groups to convert as paid.
- We can Increase the conversion rate for the profile completed as well.
- It can expand the customers to phone & website activity also.
- It is better to improve the number of page views per visit by reducing the outliers.
- The no status for print media has to be reduced and will raise Yes status for print media to increase the conversion rate.